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Erythrocyte-derived extracellular vesicles and proteins associated with such vesicles as biomarkers for Parkinson's disease

Abstract

The present description relates to methods for clinically assessing Parkinson's disease in a subject using protein biomarkers of erythrocyte-derived extracellular vesicles (EEV).

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Background/Summary

RELATED APPLICATIONS (1) This application is a Continuation of U.S. patent application Ser. No. 16/484,073, which is the U.S. National Stage of International Application No. PCT/CA2018/050150, filed Feb. 9, 2018, which designates the U.S., published in English, and claims the benefit of U.S. Provisional Application No. 62/457,350, filed Feb. 10, 2017. The entire teachings of the above applications are incorporated herein by reference.

INCORPORATION BY REFERENCE OF MATERIAL IN XML

(1) This application incorporates by reference the Sequence Listing contained in the following eXtensible Markup Language (XML) file being submitted concurrently herewith: a) File name: 54931003003_Sequence_Listing.xml; created Apr. 25, 2023, 15,256 Bytes in size. The computer readable form is incorporated herein by reference.

(2) TABLE-US-00001 SEQ Uniprot ID NO: Protein sequence of: ID 1 Axin interactor, dorsalization-associated protein Q96BJ3 2 Alpha/beta hydrolase domain-containing protein Q96IU4 14B 3 Glutamine-dependent NAD(+) synthetase E9PNF5 4 Dihydropteridine reductase

P09417 5 Alcohol dehydrogenase [NADP(+)] P14550 6 CB1 cannabinoid receptor-interacting protein 1 B8ZZB8 7 Ubiquitin carboxyl-terminal hydrolase 24 Q9UPU5 8 ATP synthase subunit alpha, mitochondrial K7EQH4

BACKGROUND

(3) Parkinson's disease (PD) is one of the most common neurodegenerative disorders affecting millions of people worldwide. Definite diagnosis for PD can only be made postmortem, for instance, by the characteristic accumulation of the protein alpha-synuclein into Lewy body inclusions observed within neurons. Currently, the diagnosis of PD is based on fitting observed symptoms and their severity into clinical rating scales such as the Unified Parkinson's Disease Rating Scale (UPDRS) or the Hoehn & Yahr scale. Current clinical assessments are subjective, however, and would benefit from improved methods of clinically assessing PD, particularly at early stages of the disease when therapeutic options are likely to be most efficient.

SUMMARY

(4) The present description relates to Parkinson's disease. More particularly, the present description relates to extracellular vesicles originating from erythrocytes, and protein biomarkers associated erythrocyte-derived extracellular vesicles, for use in clinically assessing Parkinson's disease in a subject.

(5) The present description relates to the proteomic analysis of erythrocyte-derived extracellular vesicles (EEV) from healthy control subjects, subjects with mild Parkinson's disease, and subjects with moderate Parkinson's disease, as well as the identification of protein biomarkers for which expression within EEV was found to be significantly modified therebetween. Accordingly, the present description generally relates to methods of clinically assessing Parkinson's disease based on the detection and/or quantification of the expression level of one or more protein biomarker(s) within EEV.

(6) In some aspects, the present description may relate to one or more of the following items: 1. An in vitro method for clinically assessing Parkinson's disease, the method comprising obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject having or suspected of having Parkinson's disease; and determining the expression level(s) of one or more protein biomarker(s) in the preparation of isolated EEV, wherein the one or more protein biomarker(s) comprises: (i) Alpha/beta hydrolase domain-containing protein 14B, if the subject has or is suspected of having mild or moderate Parkinson's disease; (ii) Alcohol dehydrogenase [NADP(+)], if the subject has or is suspected of having mild Parkinson's disease; (iii) ATP synthase subunit alpha, mitochondrial, if the subject has or is suspected of having moderate Parkinson's disease; or (iv) any combination of (i) to (iii), wherein the expression level of one or more of the protein biomarker(s) is indicative of, or correlates with, the subject's Parkinson's disease state. 2. An in vitro method for clinically assessing Parkinson's disease, the method comprising: obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject having or suspected of having Parkinson's disease; and determining the expression level(s) of one or more protein biomarker(s) in the preparation of isolated EEV, wherein the expression level of one or more of the protein biomarker(s) is indicative of, or correlates with, the subject's Parkinson's disease state. 3. The method of item 2, wherein the one or more protein biomarker(s) comprises at least one of: (a) Axin interactor, dorsalization-associated protein; (b) Alpha/beta hydrolase domain-containing protein 14B; (c) Glutamine-dependent NAD(+) synthetase; (d) Dihydropteridine reductase; (e) Alcohol dehydrogenase [NADP(+)]; (f) CB1 cannabinoid receptor-interacting protein 1; (g) Ubiquitin carboxyl-terminal hydrolase 24; and (h) ATP synthase subunit alpha, mitochondrial. 4. The method of item 3, wherein the one or more protein biomarkers comprises at least two of (a) to (h). 5. The method of item 3, wherein the one or more protein biomarkers comprises at least three of (a) to (h). 6. The method of item 3, wherein the one or more protein biomarkers comprises at least four of (a) to (h). 7. The method of item 3, wherein the one or more protein biomarkers comprises at least five of (a) to (h).

8. The method of item 3, wherein the one or more protein biomarkers comprises at least six of (a) to (h). 9. The method of item 3, wherein the one or more protein biomarkers comprises at least seven of (a) to (h). 10. The method of any one of items 2 to 9, wherein the one or more protein biomarkers comprises (a). 11. The method of any one of items 2 to 10, wherein the one or more protein biomarkers comprises (b). 12. The method of any one of items 2 to 11, wherein the one or more protein biomarkers comprises (c). 13. The method of any one of items 2 to 12, wherein the one or more protein biomarkers comprises (d). 14. The method of any one of items 2 to 13, wherein the one or more protein biomarkers comprises (e). 15. The method of any one of items 2 to 14, wherein the one or more protein biomarkers comprises (f). 16. The method of any one of items 2 to 15, wherein the one or more protein biomarkers comprises (g). 17. The method of any one of items 2 to 16, wherein the one or more protein biomarkers comprises (h). 18. The method of any one of items 2 to 17, further comprising determining the expression level(s) of one or more of the EEV protein(s) listed in Table 4, wherein the one or more EEV protein(s) does not comprise (a) to (h). 19. The method of any one of items 2 to 18, further comprising normalizing the expression level(s) of the one or more protein biomarker(s) to one or more of the EEV protein(s) as defined in item 18. 20. The method of any one of items 1 to 19, wherein the preparation of isolated EEV is obtained after inducing the calcium-dependent production of EEV from activated erythrocytes in the blood sample of the subject. 21. The method of any one of items 1 to 20, wherein the preparation of isolated erythrocyte-derived extracellular vesicles (EEV) is obtained by separating the EEV by flow cytometry, differential centrifugation, nanomembrane ultrafiltration, immunoabsorbent capture, size-exclusion chromatography, ultracentrifugation, magnetic activated cell sorting (MACS), nanoparticle tracking analysis, light scattering, electrophoretic light scattering, dynamic light scattering, electron microscopy, or any combination thereof. 22. The method of any one of items 1 to 21, wherein the EEV are CD235a⁺ extracellular vesicles. 23. The method of any one of items 1 to 22, wherein the EEV are TSG101⁺, Rabs⁺, CD9⁺, CD63⁺, CD81⁺, or any combination thereof. 24. The method of any one of items 1 to 23, wherein the EEV are between about 20 nm and about 1000 nm in diameter. 25. The method of any one of items 1 to 24, wherein said EEV are greater than about 100 nm in diameter. 26. The method of any one of items 1 to 25, further comprising removing hemoglobin from the preparation of isolated EEV prior to determining the expression level(s) of the one or more protein biomarker(s). 27. The method of any one of items 1 to 26, wherein determining the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV comprises contacting the protein biomarker with an antibody directed against the protein biomarker. 28. The method of any one of items 1 to 27, wherein determining the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV comprises mass spectrometry. 29. The method of item 28, wherein determining the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV comprises nano liquid chromatography tandem mass spectrometry (nanoLC-MS/MS). 30. The method of any one of items 1 to 29, wherein clinically assessing Parkinson's disease comprises diagnosing Parkinson's disease. 31. The method of any one of items 1 to 30, wherein clinically assessing Parkinson's disease in the subject comprises staging Parkinson's disease. 32. The method of any one of items 1 to 31, wherein clinically assessing Parkinson's disease comprises monitoring the progression of Parkinson's disease. 33. The method of any one of items 1 to 32, wherein clinically assessing Parkinson's disease comprises monitoring the effectiveness of treatment of a Parkinson's disease subject. 34. The method of item 32 or 33, comprising determining the expression level(s) of one or more of the protein biomarker(s) in a preparation of isolated EEV from a further blood sample of the subject obtained at a later point of time. 35. A method for analyzing proteins of erythrocyte-derived extracellular vesicles (EEV), the method comprising: (i) isolating erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject and forming an EEV preparation therefrom; (ii) extracting proteins from the EEV preparation; (iii) removing hemoglobin from the EEV preparation to obtain a hemoglobin-free EEV preparation; and (iv)

detecting EEV proteins present in the hemoglobin-free EEV preparation. 36. The method of item 35, wherein: step (iii) comprises removing a hemoglobin-containing fraction of proteins from the EEV preparation to obtain a hemoglobin-depleted EEV preparation and a hemoglobin-rich fraction; and step (iv) comprises detecting EEV proteins present in the hemoglobin-free EEV preparation and EEV proteins present in the hemoglobin-rich fraction. 37. The method of item 35 or 36, wherein the EEV proteins detected comprise the protein biomarkers as defined in any one of items 3 to 17. 38. The method of any one of items 35 to 37, wherein the preparation of isolated EEV is obtained after inducing the calcium-dependent production of EEV from activated erythrocytes in the blood sample of the subject. 39. The method of any one of items 35 to 38, wherein the preparation of isolated erythrocyte-derived extracellular vesicles (EEV) is obtained as defined in item 21. 40. The method of any one of items 35 to 39, wherein the EEV are as defined in any one of items 21 to 24. 41. The method of any one of items 35 to 40, wherein the EEV proteins are detected as defined in any one of items 26 to 28. 42. Use of one or more erythrocyte-derived extracellular vesicle (EEV) protein biomarker(s) as defined in any one of items 3 to 17 for clinically assessing Parkinson's disease in a subject. 43. The use of item 42, wherein the clinical assessment is as defined in any one of items 30 to 33. 44. A method of treating a subject with Parkinson's disease, the method comprising: (i) clinically assessing the subject according to the method of any one of items 1 to 41; and (ii) beginning or modifying the subject's Parkinson's disease treatment based on the clinical assessment in (i).

General Definitions

(7) Headings, and other identifiers, e.g., (a), (b), (i), (ii), etc., are presented merely for ease of reading the specification and claims. The use of headings or other identifiers in the specification or claims does not necessarily require the steps or elements be performed in alphabetical or numerical order or the order in which they are presented.

(8) The use of the word “a” or “an”, when used in conjunction with the term “comprising” in the claims and/or the specification may mean “one” but it is also consistent with the meaning of “one or more”, “at least one”, and “one or more than one”.

(9) The term “about” is used to indicate that a value includes the standard deviation of error for the device or method being employed to determine the value. In general, the terminology “about” is meant to designate a possible variation of up to 10%. Therefore, a variation of 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10% of a value is included in the term “about”. Unless indicated otherwise, use of the term “about” before a range applies to both ends of the range.

(10) As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, un-recited elements or method steps.

(11) Other objects, advantages and features of the present description will become more apparent upon reading of the following non-restrictive description of specific embodiments thereof, given by way of example only with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the appended drawings:

(2) FIGS. 1A-1F. Optimization of EV detection: controls for flow cytometry. FIG. 1A: To properly set the EV gate, fluorescent silica beads of 100 nm (Red), 500 nm (Blue) and 1000 nm (Yellow) were acquired on a flow cytometer Canto II modified with a FSC-PMT small particles option. The EV gate was used throughout the experiments. FIG. 1B: Serial dilutions (1, 2, 4 and 10) of

erythrocyte-derived EV (EEV) to confirm the linearity of the quantification. FIG. 1C: FSC-PMT/SSC gates of platelet-free plasma (PFP) stained with annexin V and respective fluorochrome-conjugated antibodies directed against erythrocyte (CD235a+), endothelial (CD31+/CD41-)/platelets (CD41+) and leukocytes (CD14+CD45+, monocytes; CD15+CD45+, granulocytes)-derived EV. Controls for EV labeling. FIG. 1D: Treatment with the ion chelator EDTA inhibited the binding of annexin V to phosphatidylserine. FIG. 1E: Minimal background was observed using antibodies in absence of PFP. This background was subtracted from all subsequent EV quantifications. FIG. 1F: EV sensitivity to 0.5% Triton™ was assessed. Abbreviations: AnnV, annexin V; FSC PMT-H, forward scatter photomultiplier; PBS, phosphate buffered saline; PFP, platelet free plasma; SSC-H, side scatter.

(3) FIGS. 2A and 2B. EEV: A biomarker of PD state. FIG. 2A shows the correlations between the number of erythrocyte-derived extracellular vesicles (EEV; expressed as CD235a+ EV/total number of erythrocytes) and the Unified Parkinson's Disease Rating Scale (UPDRS) of subjects (n=20). Robust correlations between the number of EEV/total number of erythrocytes and UPDRS scores (PD, n=20), displayed a clear split between mild and moderate patients, supporting the potential of EEV as a biomarker for disease state. Additional information on five patients (identified as 1-5) derived from each correlation is provided (number of EEV/total number of erythrocytes and levodopa dose equivalent) to illustrate that levodopa dosing cannot account for differences in EEV counts. Note that patient no. 5, which falls outside the confidence boundary, is the only patient on a regimen of anti-inflammatory drugs to manage arthritis. FIG. 2B shows the results of a similar analysis as in FIG. 2A, but performed on Huntington's disease subjects (n=42) using the Unified Huntington's Disease Rating Scale (UHDRS). The numbers of EEV are expressed as CD235a+ EV/total number of erythrocytes. No statistically significant correlations were found between the number of EEV/total number of erythrocytes and UHDRS scores. This argues in favor of EEV being a specific biomarker of PD. Distributions were determined using unpaired t-test with Welch's correction (PD) or one-way ANOVA (HD). Correlations were determined using Pearson's correlation, *p<0.05. Abbreviations: CD235a, glycophorin A; CTRL, Controls; EEV, erythrocyte-derived extracellular vesicle; EV, extracellular vesicle; HD, Huntington's disease; LEDD, Levodopa equivalent daily dose; PD, Parkinson's disease; Pre-HD, Pre-manifest; UHDRS, Unified Huntington's Disease Rating Scale; UPDRS, Unified Parkinson's Disease Rating Scale.

(4) FIGS. 3A and 3B. Detection of normal and phosphorylated α -Syn in EEV. FIG. 3A: Representative scanning electron microscopy observations of resting and activated erythrocytes (treated with calcium ionophore A23187 to generate EEV) in both PD patients and healthy sex- and age-matched CTRL. Scale bar: 2 μ m. FIG. 3B: Representative transmission electron microscopy images of immunogold labeling for α -Syn and α -Syn pS129 in activated erythrocytes and EEV (some examples delineated by dotted lines). Arrowheads point to positive immunolabeling for either α -Syn or α -Syn pS129. Scale bar: 100 nm. FIG. 3C: Quantification of α -Syn in EEV as detected by transmission electron microscopy and expressed as the percentage of EEVs positive for α -Syn/total number of EEV in healthy sex- and age-matched CTRL and PD patients (n=100 erythrocytes sampled in n=3 CTRL and n=3 PD). FIG. 3D: Quantification of α -Syn in EEV by ELISA assay in healthy sex- and age-matched CTRL, mild and moderate stage patients selected according to their H&Y stage (n=4 erythrocytes per group; n=13 EEV per group) revealing the absence of measurable changes in α -Syn levels between PD and healthy sex- and age-matched CTRL. Statistical analyses were performed using a Mann-Whitney U test (FIG. 3C) or a Kruskal-Wallis ANOVA (FIG. 3D). Abbreviations: α -Syn, α -synuclein; α -Syn pS129, α -synuclein phosphorylated Serine 129; CTRL, Control; EEV, erythrocyte-derived extracellular vesicle; H&Y, Hoehn and Yahr; PD, Parkinson's disease.

(5) FIGS. 4A-4D. Specific protein signature of EEV in PD patients. FIG. 4A: NanoLC-MS/MS Label-free analysis of EEV in PD patients and healthy age-matched CTRL (PD, n=4; CTRL, n=4)

revealed a total of 818 proteins, with 8 of which the expression was significantly modified as a function of PD states. FIG. 4B: The 8 differentially expressed proteins are referenced according to the gene to which they are associated, and further separated into 3 groups in relation to their expression variations in comparison to CTRL (Group I), mild PD (Group II) or moderate PD (Group III).

(6) FIG. 4C: Normalized expression (intensity) of the proteins associated with the Group I genes AIDA, ABHD14B, and NADSYN1; the Group II genes QDPR, AKR1A1, and CNRIP1; and the Group III genes USP24 and ATP5A1. FIG. 4D: Heatmap establishing correlations between disease states and the abundance of the variable proteins. Cold (C) and hot (H) colors represent low and high correlation levels, respectively. The AU p value is indicated for each node. Protein modulation was determined by unpaired t-tests with Welch's correction using the criteria of a p value under 0.05 and a minimum of 2-fold change between groups, * $p < 0.05$, ** $p < 0.01$. Abbreviations: AU, Approximately Unbiased; CTRL Control; PD, Parkinson's disease.

(7) FIGS. 5A-5C. Confirmation of EEV proteins selectively modified in PD patients by Volcano plots. The protein ratios ($\log_2(\text{ratio})$) of the three comparison (FIG. 5A: mild PD/CTRL, FIG. 5B: moderate PD/CTRL, and FIG. 5C: moderate PD/mild PD) were plotted over the corresponding Welch's test p value ($-\log_{10}(\text{p-value})$). The graphs display a V shape, as expected, and only the proteins falling outside the limits of a p value < 0.05 and absolute value of z-score > 1.96 (identified by black lines) were considered as variant proteins (arrows). Two variant proteins were excluded given that they were quantified using only one peptide. Abbreviations: CTRL: Control; PD, Parkinson's disease.

DETAILED DESCRIPTION

(8) The present description relates to the proteomic analysis of erythrocyte-derived extracellular vesicles (EEV) and the identification of protein biomarkers whose expression levels within EEV correlate with different Parkinson's disease (PD) states.

(9) In one aspect, the present description relates to a method for clinically assessing Parkinson's disease in a human subject, based on the expression level of one or more protein biomarker(s) that may correlate with or be indicative of the subject's Parkinson's disease state. As used herein, the expression “clinically assessing” or “clinical assessment” in the context of PD refers to an evaluation of a subject's PD state, which may or may not occur in a clinical setting, and which may or may not be performed by a health care professional. For example, clinically assessing may comprise screening and/or diagnosing PD in a subject having or suspected of having PD, staging a subject's PD, monitoring the progression of PD in a subject, monitoring the effect of PD medication or treatment (e.g., over time), or any combination thereof. The use of the methods described herein with other methods for clinically assessing PD subjects is also envisaged.

(10) In some aspects, the methods described herein may comprise obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject having or suspected of having Parkinson's disease. As used herein, the expression “extracellular vesicles” (EV) refers to subcellular membrane vesicles found in the extracellular environment (e.g., bodily fluids) that originate from cells, and which range in size from about 20 nm to about 1000 nm. EV may comprise exosomes, microvesicles (MV), multivesicular endosomes (MVE), or vesicles produced by apoptotic bodies, or any combination thereof, as well as other types of extracellular vesicles. Whereas the majority of the circulating EV that are detected by flow cytometric assays are likely to be MV, we do not completely exclude the potential contribution of larger exosomes or vesicles produced by apoptotic bodies. In some embodiments, the EV of the present description comprise vesicles between about 30, 40, 50, 60, 70, 80, 90, or 100 nm to about 500, 600, 700, 800, 900, or 1000 nm in size. In some embodiments, the EV of the present description comprise vesicles from 100 nm to 1000 nm in size. In some embodiments, the EV of the present description comprise vesicles between 150 nm to 1000 nm in size. All EV are composed of membrane proteins and lipids, as well as cytoplasmic components of the cell from which they

originate, such as mRNA and miRNA, organelles or infectious particles (e.g., prions, virus). A variety of methods may be used to determine the origin of EV. For example, cell surface markers (e.g., with immunolabeling and/or flow cytometry techniques) may be used to identify, enrich/purify/isolate, and/or quantify EV according to their cell of origin. Examples of such markers include: CD235a+ (erythrocytes), CD31+/CD41-(endothelial cells), CD41+ (platelets), CD45+ (leukocytes), CD45+ CD14+ (monocytes), and CD45+ CD15+(granulocytes). Of particular interest for the present description are markers that are present in (or specific for) EEV that may be used to identify, enrich/purify/isolate, and/or quantify EEV from other types of EV. Examples of such EEV markers include endosome or membrane-bonding proteins such as TSG101 and Rabs (enriched in exosomes), tetraspanins such as CD9, CD63 and CD81 (enriched in exosomes), golgi and mitochondrial proteins (enriched in MVs and absent in exosomes) (Lotvall et al., 2014). In some embodiments, the EEVs of the present description may comprise one or more of the EEV protein(s) listed in Table 4.

(11) As used herein, the expression “[marker]+EV” or “[marker]-positive” in relation to extracellular vesicles refers to the presence or detectability of that marker in an EV population of interest, regardless of whether that marker is actually detected (e.g., using an immunolabel). Conversely, the expression “[marker]-EV” or “[marker]-negative EV” refers to the absence or lack of detectability of that marker in an EV population of interest, regardless of whether that marker is actually detected (e.g., using an immunolabel). For example, the expression “CD235+EV” or “CD235a-positive EV” means EV that comprise the marker CD235a (Glycophorin A).

(12) As used herein, the term “protein biomarker” refers to a molecular indicator that is a polypeptide or protein that is associated with a particular pathological or physiological state (e.g., PD disease state). For example, the expression “Parkinson's disease biomarker” or “PD biomarker” refers to a molecular indicator that is associated with the presence, stage, and/or progression of PD in a subject. Furthermore, the term “EEV protein biomarker” refers to a protein biomarker that is expressed in EEV, whose level of expression is associated with a particular pathological or physiological state (e.g., PD disease state).

(13) In some embodiments, obtaining a preparation of isolated EEV from a blood sample of a subject (e.g., a subject having or suspected of having Parkinson's disease) may involve identifying, enriching/purifying/isolating, and/or quantifying EEV in a blood sample from the subject. In some embodiments, the blood samples may be processed to obtain platelet-free plasma (PFP), and the preparation of isolated EEV may be prepared from PFP. As used herein, the terms “enriched”, “purified”, “isolated” and the like, refer to either removing contaminants from a biological sample and/or increasing the concentration of an analyte of interest (e.g., EEV) in the sample, to an extent that is not found in nature. In some embodiments, identifying, enriching/purifying/isolating, and/or quantifying EEV may involve flow cytometry, differential centrifugation, nanomembrane ultrafiltration, immunoabsorbent capture, size-exclusion chromatography, ultracentrifugation, magnetic activated cell sorting (MACS), nanoparticle tracking analysis, light scattering, electrophoretic light scattering, dynamic light scattering, electron microscopy or any combination thereof, or using other techniques that can separate vesicles based on their size and/or surface protein expression. Quantifying EEVs may also be performed by methods such as nanoparticle tracking (NTA), biochemical approaches and semi-quantitative electron microscopy approaches. In some embodiments, the methods described herein may further comprise quantifying the level of EEV in a blood sample from a subject. The quantification of EEV may be expressed as a relative value by normalizing the number of EEV (e.g., in terms of the total number of erythrocytes).

(14) In some embodiments, preparations of isolated EEV described herein may comprise at least 70%, 75%, 80%, 85%, 90%, 95%, or 99% (e.g., by weight or number) of EEV, and/or less than 30%, 25%, 20%, 15%, 10%, 5%, or 1% (e.g., by weight or number) of non-EEV.

(15) In some embodiments, the methods described herein may comprise obtaining a blood sample from a subject that is less than 5 mL, less than 4 mL, less than 3 mL, less than 2 mL, less than 1

mL, less than 900 μ L, less than 800 μ L, less than 700 μ L, less than 600 μ L, less than 500 μ L, less than 400 μ L, less than 300 μ L, less than 200 μ L, less than 100 μ L, less than 50 μ L, less than 40 μ L, less than 30 μ L, less than 25 μ L, less than 20 μ L, or less than 10 μ L.

(16) In some embodiments, the methods described herein may comprise detecting or determining the expression level of an EEV protein biomarker that is differentially expressed in controls (e.g., non-PD subjects) and/or in different PD states (e.g., mild, moderate, or severe), for example based on the unified Parkinson's disease rating scale (UPDRS). As used herein, the expression "mild PD patients" are defined as patients characterized by a UPDRS score of lower than 37; "moderate PD patients" are defined as patients characterized by a UPDRS score of between 37 and and "severe PD patients" are defined as patients characterized by a UPDRS score of greater than 75. In some embodiments, the EEV protein biomarkers described herein may be used, for example, to distinguish between mild, moderate and severe PD patients. In some embodiments, the EEV protein biomarkers described herein may be used to distinguish between patients characterized by UPDRS scores within different ranges from those mentioned above. As used herein, "control subjects" or "controls" refer to non-PD subjects (e.g., healthy subjects).

(17) In some embodiments, the methods described herein comprise determining the expression level(s) of one or more protein biomarker(s) in a preparation of isolated EEV, wherein the protein biomarker(s) is/are one or more of the proteins defined in the Table below.

(18) TABLE-US-00002 SEQ ID Related Related Corresponding Groups Protein biomarker NO: accession no. UniProt ID gene name I Axin interactor, dorsalization- 1 NP_073742.2 Q96BJ3 AIDA (Control) associated protein Alpha/beta hydrolase domain- 2 NP_116139.1 Q96IU4 ABHD14B containing protein 14B Glutamine-dependent NAD(+) 3 EAW74792.1 E9PNF5 NADSYN1 synthetase II Dihydropteridine reductase 4 NP_001293069.1 QDPR QDPR (mild PD) Alcohol dehydrogenase [NADP(+)] 5 AAP36383.1 P14550 AKR1A1 CB1 cannabinoid receptor-interacting 6 NP_056278.1 B8ZZB8 CNRIP1 protein 1 NP_001104571.1 III Ubiquitin carboxyl-terminal hydrolase 7 NP_056121.2 Q9UPU5 USP24 (moderate 24 PD) ATP synthase subunit alpha, 8 EAX01470.1 K7EQH4 ATP5A1 mitochondrial

(19) In some embodiments, the protein biomarkers identified herein as belonging to Group I may be differentially expressed in the EEV of healthy or non-PD control subjects, as compared to those of mild and/or moderate PD subjects. In some embodiments, the protein biomarkers identified herein as belonging to Group II may be differentially expressed in the EEV of mild PD subjects, as compared to those of healthy or non-PD control subjects and/or moderate PD subjects. In some embodiments, the protein biomarkers identified herein as belonging to Group III may be differentially expressed in the EEV of moderate PD subjects, as compared to those of healthy or non-PD control subjects and/or mild PD subjects.

(20) In some embodiments, the methods described herein may comprise determining the expression of one or more of the protein biomarkers described herein, based on the PD state which is to be clinically assessed. For example, different protein biomarkers or combinations of protein biomarkers may be selected based on, for example, the differential expression patterns reported herein (e.g., in Table 3.1 and in FIG. 4C). In some embodiments, the methods described herein may comprise determining the expression of: (i) Alpha/beta hydrolase domain-containing protein 14B, if the subject has or is suspected of having mild or moderate Parkinson's disease; (ii) Alcohol dehydrogenase [NADP(+)], if the subject has or is suspected of having mild Parkinson's disease; (iii) ATP synthase subunit alpha, mitochondrial, if the subject has or is suspected of having moderate Parkinson's disease; or (iv) any combination of (i) to (iii). In some embodiments, the subject is determined to have mild or moderate PD when the expression level of alpha/beta hydrolase domain-containing protein 14B in the preparation of isolated EEV from a blood sample of the subject is lower than that corresponding to a control subject (i.e., not having PD). In some embodiments, the subject is determined to have mild PD when the expression level of alcohol dehydrogenase [NADP(+)] in the preparation of isolated EEV from a blood sample of the subject is

higher than that corresponding to a control subject (i.e., not having PD). In some embodiments, the subject is determined to have moderate PD when the expression level of ATP synthase subunit alpha, mitochondrial in the preparation of isolated EEV from a blood sample of the subject is higher than that corresponding to a control subject (i.e., not having PD). In some embodiments, the terms “higher” or “lower” may refer to a difference in expression from the reference value of at least 1.5-fold, 2-fold, 2.5-fold, 3-fold, or 3.5-fold. In some embodiments, methods described herein may comprise determining the expression level(s) of at least one, at least two, at least three, at least four, at least five, at least six, at least seven, or all eight of the protein biomarker(s): (a) Axin interactor, dorsalization-associated protein (represented by SEQ ID NO: 1); (b) Alpha/beta hydrolase domain-containing protein 14B (represented by SEQ ID NO: 2); (c) Glutamine-dependent NAD(+) synthetase (represented by SEQ ID NO: 3); (d) Dihydropteridine reductase (represented by SEQ ID NO: 4); (e) Alcohol dehydrogenase [NADP(+)] (represented by SEQ ID NO: 5); (f) CB1 cannabinoid receptor-interacting protein 1 (represented by SEQ ID NO: 6); (g) Ubiquitin carboxyl-terminal hydrolase 24 (represented by SEQ ID NO: 7); and (h) ATP synthase subunit alpha, mitochondrial (represented by SEQ ID NO: 8).

(21) In some embodiments, the expression level of one or more of the protein biomarker(s) described herein may be determined by detecting and/or quantifying the presence of a polypeptide fragment of any one of the polypeptides of SEQ ID NOs: 1-8. Such fragments may comprise or consist of at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 45, 50, 55, 60, 65, 70, 75, 80, 90, 95, 100, 125, 150, 175, 200, or more contiguous amino acids of any one of SEQ ID NOs: 1-8. In some embodiments, the polypeptides of any one of SEQ ID NOs: 1-8 may be present in the EEV (or a preparation of isolated EEV) as a fragment of SEQ ID NOs: 1-8. Detection of such fragments are considered within the scope of the present description. In some embodiments, a protein biomarker described herein may comprise one or more fragments of any one of SEQ ID NOs: 1-8, for example a fragment comprising or consisting of: residues 127-192 of SEQ ID NO: 1; residues 188-200 of SEQ ID NO: 2; residues 112-260 or residues 349-446 of SEQ ID NO: 3; residues 1-213 of SEQ ID NO: 4; residues 1-325 of SEQ ID NO: 5; residues 1-110 or residues 84-110 of SEQ ID NO: 6; residues 2570-2620 of SEQ ID NO: 7; or residues 51-161, residues 111-137, or residues 53-152 of SEQ ID NO: 8.

(22) In some embodiments, methods described herein may comprise determining the expression level(s) of at least one, at least two, at least three, at least four, at least five, at least six, at least seven, or all eight of the protein biomarker(s) encoded by the human genes AIDA, ABHD148, NADSYN1, QDPR, AKR1A1, CNRIP1, USP24, and ATP5A1.

(23) In some embodiments, methods described herein may comprise determining the expression level(s) of at least one, at least two, at least three, at least four, at least five, at least six, at least seven, or all eight of the protein biomarker(s) defined by accession numbers: NP_073742.2, NP_116139.1, EAW74792.1, NP_001293069.1, AAP36383.1, NP_056278.1, NP_001104571.1, NP_056121.2, and EAX01470.1.

(24) In some embodiments, protein biomarker (a) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 1. In some embodiments, protein biomarker (b) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 2. In some embodiments, protein biomarker (c) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 3. In some embodiments, protein biomarker (d) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 4. In some embodiments, protein biomarker (e) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 5. In some embodiments, protein biomarker (f) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 6. In some embodiments, protein biomarker (g) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of

SEQ ID NO: 7. In some embodiments, protein biomarker (f) may be at least 70%, 75%, 80%, 85%, 90%, or 95% identical to the amino acid sequence of SEQ ID NO: 8.

(25) In some embodiments, the methods described herein may comprise determining the expression levels of an EEV protein signature comprising any combination of the protein biomarkers (a) to (h). In some embodiments, the methods described herein may comprise determining the expression levels of at least one of protein biomarkers (a), (b), and (c) [Group I]; at least one of protein biomarkers (d), (e), and (f) [Group II]; and/or at least one of protein biomarkers (g) and (h) [Group III].

(26) In some embodiments, the methods described herein may further comprise comparing the expression level of the protein biomarker(s) so detected and comparing the expression level to a reference value corresponding to that of a control group, non-PD group, mild PD group, moderate PF group, or severe PD group.

(27) In some embodiments, the EEV protein biomarkers described herein do not comprise alpha-synuclein; and/or the methods described herein do not comprise detecting and/or quantifying alpha-synuclein.

(28) In some embodiments, the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV may comprise contacting the protein biomarker with an antibody directed against the protein biomarker (e.g., an immunoassay). In some embodiments, the antibody specifically binds to the polypeptide of any one of SEQ ID NOs: 1-8. Antibodies against the proteins of SEQ ID NOs: 1-8 are commercially available, for example from Novus Biologicals (e.g., catalog numbers NBP1-88323, NBP2-26122, H00055191-M01, H00005860-M02, NBP2-02164, NBP1-86800, NB100-40830, and NBP2-38525, respectively). In some embodiments, antibodies described herein may bind to, or be raised against, a fragment of any one of SEQ ID NOs: 1-8, for example a fragment comprising or consisting of: residues 127-192 of SEQ ID NO: 1; residues 188-200 of SEQ ID NO: 2; residues 112-260 or residues 349-446 of SEQ ID NO: 3; residues 1-213 of SEQ ID NO: 4; residues 1-325 of SEQ ID NO: 5; residues 84-110 of SEQ ID NO: 6; residues 2570-2620 of SEQ ID NO: 7; or residues 53-152 of SEQ ID NO: 8. In some embodiments, antibodies described herein may bind to an epitope comprising or consisting of at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 contiguous residues of any one of SEQ ID NOs: 1-8.

(29) As used herein, the term “antibody” may encompass any type of antibody, including but not limited to monoclonal antibodies, polyclonal antibodies, “antigen-binding fragments” (or portion), such as Fab, Fab', F(ab')₂, Fd, Fv, Fc, etc., of intact antibodies that retain the ability to specifically bind to a given antigen (e.g., an EEV protein described herein), an isolated complementarity determining region (CDR), bispecific antibodies, heteroconjugate antibodies, mutants thereof, fusion proteins having an antibody, or antigen-binding fragment thereof, (e.g., a domain antibody), single chain (ScFv) and single domain antibodies (e.g., shark and camelid antibodies), maxibodies, minibodies, intrabodies, diabodies, triabodies, tetrabodies, v-NAR and bis-scFv, humanized antibodies, chimeric antibodies and any other modified configuration of the immunoglobulin molecule that includes an antigen recognition site of the required specificity, including glycosylation variants of antibodies, amino acid sequence variants of antibodies, and covalently modified antibodies. The antibodies may be murine, rat, human, or any other origin (including chimeric or humanized antibodies).

(30) In some embodiments, the expression level(s) of one or more of the protein biomarker(s) in the preparation of isolated EEV may comprise mass spectrometry (e.g., nano liquid chromatography tandem mass spectrometry (nanoLC MS/MS)).

(31) In some embodiments, the methods described herein may comprise combining the expression of one or more of the protein biomarker(s) in the preparation of isolated EEV with the quantity of EEV to increase the power of a biomarker described herein.

(32) In some embodiments, the methods described herein may further comprise determining the

expression level(s) of one or more of the EEV protein(s) listed in Table 4 (Proteins identified in the EEV proteome), wherein the one or more EEV protein(s) does not comprise (a) to (h). The expression level of one or more EEV proteins that are not differentially expressed between PD states may be useful for example as normalization parameters in conjunction with the one or more of the protein biomarkers (a) to (h).

(33) In some embodiments, the methods described herein may comprise determining the expression level(s) of one or more of the EEV protein(s) defined by UniProt ID: P02549, P11277, P16157, P02730, P55072, P16452, P11171, P04040, Q13228, B4DT77, Q8WUM4, P35612, P68871, Q00610, P69905, B4DVE7, J3QLD9, P08758, P09525, O75955, P11142, P32119, P00491, P27105, P00918, P23634, C9JIF9, P00915, P63261, Q5VU58, P30041, E7EU23, Q00013, E7EV01, J3KPS3, Q08495, P23276, P69892, P60174, P62258, O75326, O75340, E7EV99, Q5VZU9, P53396, C9J0K6, P04406, P07738, B7Z319, Q86X55, P07195, P23526, Q32Q12, B7Z7A9, P11166, P22303, Q9NP58, P40925, P00352, F2Z2V0, K7EMC9, F5H7S3, A6NN80, P30043, H7BXD5, P04083, P62937, P37837, Q06830, O75131, P00390, E7EQB2, P07384, P02042, P30086, P35613, Q9H0U4, P63092, P48506, P06702, Q9UBV8, P17931, P28066, P07451, E7EQ12, P50895, P28074, G3V5Z7, P25786, G3V1D3, P49247, Q5T9B7, P25789, B4E022, J3QS39, HOY7A7, P28070, Q9H4G4, Q9BY43, P48426, P28289, P07911, Q9GZP4, P78417, P25788, O14818, P08107, H0YD13, P61225, P05109, P23528, Q99808, P84077, P31946, C9JIS1, P53990, Q99497, F5H7U0, B7Z7E9, P62834, P04899, P25325, Q9NP79, P00492, Q16531, P22314, P00441, H7BY58, P10768, P09543, P06733, P26038, O75368, K7EQ48, P25787, P49721, F5H8J2, C9J9P4, P09211, B5MDF5, P07900, K7EQ02, Q13630, F5H0T1, P50502, P20618, P62805, P51148, H7C2G2, J3KQ18, H3BPK3, B4DIT7, O43633, B4DQH4, Q9UN37, I3L397, Q9Y5Z4, Q9UKV8, F5H442, H3BLV0, P02008, E7EPV7, P63104, P02724, U3KQE2, Q9NP59, Q5QPM9, C9J8U2, P61981, Q9UQ80, E5RJR5, Q9NRV9, Q9H444, P10599, Q5VSJ9, Q9UK41, Q9Y3I1, P61026, Q14974, P27797, P18669, P54725, P30613, P63000, P05164, F5GWY2, P48637, F5H5V4, G5E9R5, P46976, P28072, P26447, F5GXQ0, P08754, Q99436, P62942, U3KQK0, J3QKR3, P01116, P13489, Q08722, Q5T123, Q8WYQ7, O75695, P00167, Q9Y4D1, P11021, H7C1D4, P07737, M0R389, A6NJA2, P10644, Q9BS40, G5EA52, P53004, Q04656, H9KV70, O00299, F8WF69, G3V2F7, F8WDS9, P60891, K7ESE8, H0YNE3, P16930, F8VSD4, P07203, P62328, E5RIW3, M0R0Y2, P15374, P04921, H0YDI1, B4E220, C9JEN3, F5H2R5, Q53TN4, Q9NZD4, Q8NHG7, Q5JYX0, Q71RC9, E9PNW4, P09105, R4GN98, O75531, Q5T6W5, F5H4Q5, J3QK90, H3BV85, Q9NRX4, H3BS66, E7ESC6, P68402, Q9BRF8, P08246, E9PN50, E7EUC7, B8ZZB8, E9PCS3, P59666, O15400, P00338, P61970, E7EMV0, F5GY90, P61020, Q99828, B4DUA0, C9JTY3, P27348, H0YKZ7, P08238, J3KQP6, A6NMU3, P53985, F6USW4, O14964, P20020, P36959, Q9Y376, Q9Y6M5, Q81Z83, Q99459, P06132, J3KNT0, P49189, H3BNT7, P05023, P34932, K7EMV3, Q8IU68, E7ENZ3, Q5TZA2, Q9P203, Q7LBR1, U3KQ56, H0YJ11, Q9UDT6, P09960, Q5HY54, E9PJL5, G3V2U7, E9PQN4, Q9BSL1, Q04917, B7ZBP9, Q8NDC0, P68133, P69891, S4R3Y4, I3L3E4, Q16570, Q5VY30, E7END7, Q5VU59, P17066, Q04760, D6RD66, K7EM02, P14209, E9PIR7, K7EMQ9, P15531, H7BZT4, O00560, Q9BVM4, K7EKH5, P49773, H0YBY6, Q9Y624, B1AKQ8, K7EKN6, I3L0K2, A8MXY0, O14773, E9PNW0, Q5TDH0, Q96JM4, F5GWT9, F2Z3J2, J3QL74, E9PJC7, Q9H936, D6RD63, Q6B0K9, Q31611, H7BY04, Q9UL25, H7C3P7, P08311, E9PE37, G3V1N2, P00387, O75339, P14324, K7EKG2, P02549, P11277, P16157, P55072, P111714, P35579, Q8WUM4, P02730, P16452, P04040, A0A087WVQ6, P35612, P16157, Q14254, P20073, O75955, P53396, P06753, P49368, P236344, P11142, P60709, Q5T4S7, P78371, P28289, P50395, P27105, P68871, P02730, Q00013, J3KPS3, P00352, Q86VP6, P49327, Q13228, P00915, P50991, P50990, P32119, P50995, P69905, P00558, H7BXX9, P08758, P09525, P07900, P48643, P07384, P22314, P04406, Q08495, Q99832, P29144, P30041, E9PM69, P40227, P50570, E7EQB2, P31948, E7ESC6, E7EV99, P62258, P30613, A0A0G2JIW1, Q16531, P11021, F5H2F4, P07195, P45974, O43242, C9J0K6, P08133, Q13200, P23276, P34932, A0A0A0MSI0, Q9Y230, Q5XPI4,

P68871, P60174, P00491, C9JIF9, H7BYY1, P35998, P17987, P09543, Q99460, Q9Y4E8, Q90009, P26038, P04083, P30043, P11166, P00918, P06733, Q5TDH0, B0QZ18, O75326, P05164, Q9Y265, P29401, I3L0N3, Q4VB86, P11277, P13716, P07738, P48506, Q99816, O14818, P23526, P61225, O00231, P11413, P00338, Q99808, A6NJA2, Q06323, or any combination thereof.

(34) In some embodiments, the methods described herein may comprise determining the expression level(s) of one or more of the EEV protein(s) defined by protein accession number: NP_000017.1, NP_000022.3, NP_000028.3, NP_000034.1, NP_000036.2, NP_000043.4, NP_000058.1, NP_000110.2, NP_000128.1, NP_000134.2, NP_000149.3, NP_000166.2, NP_000169.1, NP_000175.1, NP_000185.1, NP_000230.1, NP_000241.1, NP_000260.1, NP_000261.2, NP_000276.2, NP_000280.1, NP_000282.1, NP_000289.1, NP_000311.2, NP_000333.1, NP_000338.3, NP_000356.1, NP_000365.3, NP_000366.1, NP_000382.3, NP_000389.1, NP_000393.4, NP_000410.2, NP_000411.1, NP_000421.1, NP_000445.1, NP_000471.1, NP_000507.1, NP_000508.1, NP_000509.1, NP_000510.1, NP_000549.1, NP_000550.2, NP_000572.2, NP_000623.2, NP_000628.2, NP_000656.1, NP_000678.1, NP_000680.2, NP_000687.3, NP_000691.1, NP_000692.2, NP_000703.2, NP_000704.1, NP_000792.1, NP_000843.1, NP_000886.1, NP_001001323.1, NP_001001396.1, NP_001001521.1, NP_001002021.2, NP_001002857.1, NP_001002858.1, NP_001002860.2, NP_001003938.1, NP_001003945.1, NP_001005360.1, NP_001005361.1, NP_001005362.1, NP_001005386.1, NP_001005753.1, NP_001007068.1, NP_001007069.1, NP_001007070.1, NP_001007071.1, NP_001008390.1, NP_001008800.1, NP_001009185.1, NP_001009186.1, NP_001009570.1, NP_001010935.1, NP_001010942.1, NP_001013275.1, NP_001013454.1, NP_001017963.2, NP_001019397.1, NP_001019398.1, NP_001019399.1, NP_001019820.1, NP_001020029.1, NP_001020276.1, NP_001020560.1, NP_001020561.1, NP_001026897.1, NP_001026997.1, NP_001027017.1, NP_001028196.1, NP_001028690.1, NP_001028691.1, NP_001028692.1, NP_001029197.1, NP_001029249.1, NP_001030611.1, NP_001034221.1, NP_001034271.1, NP_001034288.1, NP_001034455.1, NP_001034456.1, NP_001034679.2, NP_001034680.2, NP_001034708.1, NP_001034891.1, NP_001035517.1, NP_001035784.1, NP_001035810.1, NP_001035941.1, NP_001036816.1, NP_001036817.1, NP_001036818.1, NP_001055.1, NP_001070654.1, NP_001070956.1, NP_001070957.1, NP_001070958.1, NP_001071643.1, NP_001071645.1, NP_001073379.1, NP_001077086.1, NP_001077861.1, NP_001087.2, NP_001091.1, NP_001092.1, NP_001092006.1, NP_001092925.1, NP_001096137.1, NP_001096138.1, NP_001098985.1, NP_001099000.1, NP_001107606.1, NP_001107607.1, NP_001107608.1, NP_001107609.1, NP_001107610.1, NP_001107611.1, NP_001108628.1, NP_001112362.1, NP_001116370.1, NP_001116423.1, NP_001116849.1, NP_001116850.1, NP_001118.3, NP_001120855.1, NP_001121776.1, NP_001121777.1, NP_001121778.1, NP_001122060.3, NP_001122301.1, NP_001122302.1, NP_001122303.1, NP_001123291.1, NP_001123654.1, NP_001123655.1, NP_001123989.1, NP_001124197.1, NP_001124321.1, NP_001124322.1, NP_001124384.1, NP_001128527.1, NP_001128711.1, NP_001129171.1, NP_001129172.1, NP_001129173.1, NP_001129174.1, NP_001129293.1, NP_001129294.1, NP_001129333.1, NP_001129487.1, NP_001129511.1, NP_001135827.1, NP_001135828.1, NP_001135917.1, NP_001135918.1, NP_001136336.2, NP_001137359.1, NP_001137430.1, NP_001137457.1, NP_001138404.1, NP_001138436.1, NP_001138437.1, NP_001138438.1, NP_001138439.1, NP_001138440.1, NP_001138441.1, NP_001138442.1, NP_001138443.1, NP_001138868.1, NP_001138872.1, NP_001139280.1, NP_001139281.1, NP_001139411.1, NP_001139412.1, NP_001139501.1, NP_001139502.1, NP_001139508.2, NP_001139509.1, NP_001139748.1, NP_001139786.1, NP_001144.1, NP_001145.1, NP_001146.2, NP_001147.1, NP_001148.1, NP_001152759.1, NP_001153705.1, NP_001153706.1, NP_001154840.1, NP_001155059.1, NP_001155238.1, NP_001155901.1, NP_001156467.1, NP_001156852.1, NP_001157565.1, NP_001157566.1, NP_001157567.1, NP_001158095.1, NP_001158302.1,

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(35) In some embodiments, the preparation of isolated EEV described herein may be obtained after

inducing the production of EEV from activated erythrocytes in the blood sample of the subject. Erythrocytes may be activated to induce EEV production, for example, using a calcium ionophore (e.g., A23187), cold-storage, or ATP depletion (Prudent et al., 2015).

(36) In some embodiments, the methods described herein may comprise a step of removing or depleting hemoglobin from a preparation of EEV prior to determining the level of expression of EEV proteins. Given the significant amounts of hemoglobin within erythrocytes that could mask the true nature of the protein signature in EEV, this step may improve the detection of non-hemoglobin EEV proteins. In some embodiments, hemoglobin-depleted and hemoglobin-rich fractions may be prepared in parallel and analyzed separated for expression of EEV proteins (e.g., in mass spectrometry-based proteomic detection techniques).

(37) In some aspects, the methods described herein may further comprise comparing the level of an EEV protein biomarker to a suitable reference value indicative of the presence, stage and/or progression of Parkinson's disease, thereby clinically assessing Parkinson's disease in the subject.

(38) As used herein, the expression “reference value” means a control value or range of values corresponding to a known level or range of an EEV protein biomarker associated with the presence, stage and/or progression of Parkinson's disease. In some embodiments, for example where the expression level of an EEV protein biomarker has previously been measured in a blood sample from a subject, the reference value may be a value corresponding to the same subject's previous reading (e.g., a baseline). The term “suitable” in the expression “suitable reference value” reflects the observations reported herein that the number of EEV (and/or the protein expressed therein) in blood samples from PD subjects may vary depending on, for example, factors which may also affect the EV and/or EEV levels. For example, it is reported herein that a subject's EEV levels may be affected by whether or not the subject is being treated for their PD symptoms, whether the subject has or previously had cancer, whether the subject has or previously had diabetes, or whether the subject is taking anti-inflammatory medication.

(39) In some embodiments, the present description relates to a method for treating a subject with PD, the method comprising clinically assessing Parkinson's disease in the subject by a method described herein, and commencing, administering, and/or modifying PD treatment based on the clinical assessment.

(40) The scope of the claims should not be limited by the particular embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole

EXAMPLES

Example 1—Methods

(41) 1.1 Participant Recruitment and Ethic Statement

(42) Human blood was obtained from two cohorts of participants. The first cohort was composed of Parkinson's disease (PD) patients and healthy age- and sex-matched Controls, and the second cohort was composed of Huntington's disease (HD) patients and healthy age- and sex-matched Controls. The demographics for both cohorts are shown in Table 1. For the two cohorts, the Controls were recruited amongst the caregivers, spouses, family and friends of the patients. Institutional review boards approved this study (CHU de Québec-Université Laval, #A13-2-1096; CHUM, #14.228; Cambridge Central Regional Ethics Committee, REC #03/303 & #08/H0306/26; and Cambridge University Hospitals Foundation Trust Research and Development department, R&D #A085170 & A091246) in accordance with the Declaration of Helsinki, and written informed consent was obtained from all participants.

(43) In the case of PD patients, the clinical evaluation included measures on the Unified Parkinson Disease Rating Scale (UPDRS), the Hoehn and Yahr (H&Y) stage, the Mini Mental State Examination (MMSE), the Addenbrooke's Cognitive Examination (ACE), and the Beck Depression Inventory (BDI). In the case of the HD patients, we collected their scores on the Unified Huntington Disease Rating Scale (UPDRS), Total Functional capacity (TFC) and calculated values for burden of disease (BDS). All the clinical evaluations were conducted within 9 months of the

blood drive. Participants were further asked to fill out a questionnaire related to health issues and medication and their full blood count performed on the day of blood sampling.

(44) Of note, participants excluded from the present EEV-related analyses included those with diabetes and those suffering or having suffered from cancer, because we observed a significant PD-independent increase in EEV concentration in the platelet-free plasma of these participants. Furthermore, PFP samples with elevated free hemoglobin (>45 000 ng/mL), potentially due to hemolysis at blood sampling, were also excluded from EEV-related analyses, which explains the discrepancies between the total number of participants initially recruited and those contained in each analysis.

(45) TABLE-US-00003 TABLE 1 Participant demographics Table 1: Participant clinical information. Parkinson's disease (PD) cohort PD Patients - Stages of disease Ctrl Unknown Mild Moderate Severe P value n 37 7 12 33 8 Age 66.8 69.8 66.7 71.1 75.0* 0.04 Gender F (M) 18 (19) 1 (6) 6 (6) 16 (17) 0 (8) 0.05 Disease severity Hoehn & Yahr (n) 1 ± 0.3 (12) 2 ± 0.2 (33) 3 ± 0.5 (8) <0.0001 UPDRS (n) 38 ± 11 (6) 52 ± 19 (17) 73 ± 20 (6) 0.02 ACE (n) 96 ± 4 (6) 92 ± 7 (17) 84 ± 14 (6) 0.13 MMSE (n) 29 ± 2 (7) 29 ± 1 (19) 26 ± 3 (6) 0.01 BDI (n) 3 ± 2 (6) 4 ± 2 (17) 13 ± 7 (4) 0.03 Comorbidities Asthma 3 1 1 5 0 0.71 Hypertension 10 1 2 10 3 0.76 Diabetes 2 0 0 1 2 0.10 Cancer 5 0 3 4 1 0.64 Allergies 2 0 2 6 2 0.28 Depression 3 1 2 1 2 0.29 Hypercholesterolemia 5 0 1 6 1 0.73 Huntington's disease (HD) cohort HD Patients - Stages of disease Ctrl Pre-HD Stage 1 Stage 2 Stage 3 Stage 4 Stage 5 P value n 55 11 15 13 12 10 2 Age 55.0 37.5 53.1 54.2 58.3 58.1 55.5 0.02 Gender F (M) 31 (22) 6 (5) 5 (10) 4 (9) 8 (4) 7 (3) 1 (1) 0.26 Disease severity UHDRS (n) 2.7 (11) 15.7 (14) 34.5 (11) 42.9 (12) 55.9 (10) 67.5 (2) <0.001 TFC (n) 13 (16) 13 (11) 12.5 (15) 7.8 (13) 4.3 (12) 1.6 (10) 0 (2) <0.001 CAG (n) 28.3 (3) 41.1 (10) 42.3 (13) 42.6 (12) 43.7 (7) 44.3 (7) <0.001 BDS (n) 206 (10) 337 (13) 356 (12) 442 (7) 465 (7) <0.001 Comorbidities Asthma 0 0 1 0 0 0 0 0.65 Hypertension 4 1 2 1 1 2 0 0.92 Diabetes 3 1 1 1 1 1 0 0.99 Cancer 0 0 0 0 0 0 0 0 Allergies 3 0 2 2 0 0 0 0.33 Depression 8 1 1 3 6 4 1 0.0497 Hypercholesterolemia 8 1 1 0 0 1 0 0.32 (PD cohort) Disease severity levels in relation to the H&Y scale (score): Mild (1-1.5); Moderate (2-2.5); Severe (3-3.5). *p < 0.05 vs. CTRL. Statistical analyses were performed using a Welch ANOVA followed by Dunnett's multiple comparison test. Disease severity was evaluated within 6 months of blood sampling. Comorbidities were determined from medical information reported by the participant or caregiver. Cancer refers to participant having suffered from cancer in the past. (HD cohort) Disease severity levels in relation to the TFC scale: Stage 1 (11-13); Stage 2 (7-10); Stage 3 (3-6); Stage 4 (1-2); Stage 5 (0). Disease severity was evaluated within 6 months of blood sampling. Comorbidities were determined from medical information reported by the participant or caregiver. Abbreviations: ACE, Addenbrooke's cognitive examination; BDI, Beck depression inventory; BDS, Burden of Disease Score; CAG, Trinucleotide repeat; MMSE, Mini-Mental State Examination; UHDRS, Unified Huntington's Disease Rating Scale; TFC, Total Function Capacity.

1.2 Preparation of Platelet-Free Plasma (PFP) and Extracellular Vesicle (EV) Labeling

(46) Citrated blood was centrifuged twice for 15 minutes at 2500 g at room temperature. Platelet-free plasma (PFP) was harvested and stored at -80° C. within 2 hours of collection following guidelines suggested by Lacroix and colleagues (Lacroix et al., 2012).

(47) For all experiments, diluted annexin-V buffer (BD Pharmingen, Mississauga, ON, Canada) and phosphate buffered saline (PBS) were filtered on 0.22 µm pore size membranes. To quantify the EV according to their cell of origin, the following surface markers were used: CD235a+ (erythrocytes) (5 µL), CD31+/CD41- (endothelial cells) (1 µL), CD41+ (platelets) (5 µL), CD45+ (leukocytes) (3 µL), CD45+CD14+ (monocytes) (10 µL), and CD45+CD15+ (granulocytes) (2 µL), with or without annexin-V staining (5 µL). PFP (5 µL) was incubated with Phenylalanyl-prolyl-arginyl Chloromethyl Ketone (PPACK) (Calbiochem, Etobicoke, ON, Canada) for 5 minutes, followed by a 30-minute incubation with antibodies and annexin-V in a final PBS volume of 100 µL, all at room temperature. Finally, the samples were diluted to a final volume of 2 mL prior to

FACS analysis. The following antibodies were purchased at BD Pharmingen and used throughout the experiments: FITC-conjugated mouse anti-human CD235a (clone GA-R2 (HIR2), 1/20), PE-conjugated mouse anti-human CD31 (clone WM59, 1/100), V450-conjugated mouse anti-human CD41a (clone HIPS, 1/20), APC mouse anti-human CD14 (clone M5E2, 1/10), PE-conjugated mouse anti-human CD15 (clone H198, 1/50), V450-conjugated mouse anti-human CD45 (clone H130, 1/33), V450- and PerCP-CyTM5.5-conjugated annexin-V (1/33 and 1/10, respectively).

(48) 1.3 Flow Cytometry Quantification

(49) For EV quantification, we used a FACS Canto II Special Order Research Product equipped with a forward scatter (FSC) coupled to a photomultiplier tube (FSC-PMT). Flow cytometer performance tracking was carried out daily using the BD cytometer setup and tracking beads (BD Biosciences, San Jose, CA, USA). The size of the EV was determined using fluorescent silicone beads of 100, 500 and 1000 nm. Controls and optimization of the detection method are presented in FIGS. 1A-1F. The settings for the EV detection were determined as described previously (Rousseau et al., 2015) using a threshold of 200 for SSC. Between PD and HD analyses, the blue laser had to be replaced for maintenance issues and therefore laser settings were reassessed. For FSC-PMT, the assigned voltage was 363 (PD) and 160 (HD) Volts. For SSC, the assigned voltage was 407 (PD) and 300 (HD) Volts. All other parameters were set between 450 and 500 Volts. The acquisition of EV was performed at low speed with an approximate rate of 10 μ L/min. To determine background noise level, antibody mixes were incubated in absence of PFP sample and unlabeled PFP was used as a negative control.

(50) 1.4 Statistical Analyses

(51) All statistical analyses pertaining to Examples 2.1 and 2.2 were performed using “The Statistics and Machine Learning Toolbox” provided by MathWorksTM under the MATLABTM platform. The version used was MATLAB® R2015a. The analysis included the scatter plot, the classical least-squares linear regression model, the R-squared and p values, as well as Pearson's goodness-of-fit model. Interval cut-off values were determined using a loop program developed in MATLABTM. Model diagnostics, including residual behaviour and homoscedastivity, were also obtained with the same Toolbox.

(52) 1.5 Production and Purification of EEV

(53) Blood was collected in heparin tubes and centrifuged for 10 minutes at 282 g at room temperature. Blood cells were washed first in PBS-2% FBS, then with 0.9% sodium chloride solution and centrifuged for 10 minutes at 750 g. To avoid leukocyte and/or platelet contamination, the buffy coat and the upper fraction of erythrocytes were removed. To preserve erythrocytes, two volumes of glycerolyte 57 solution (57% glycerol, 142 mM sodium lactate, 1 mM KCl, 25 mM sodium phosphate pH 6.8) were added to the pellet and stored at -80° C.

(54) For the production of EEV, red bloods cells were thawed and EV production was induced as previously described (Minetti et al., 2004). Briefly, the erythrocyte pellet was activated with 3 volumes of calcium ionophore solution (150 mM NaCl; 10 mM Tris-HCl; 1 mM CaCl.sub.2; 5 μ M ionophore A23187 (Sigma, St Louis, MO)) for 30 minutes at 37° C. The activation was stopped by the addition of 5 mM EDTA. Remaining erythrocytes were pelleted at 15 000 g for 20 minutes. The EEV were centrifuged at 20 000 g for 90 minutes and washed once in PBS. The EEV pellet was resuspended in PBS and frozen at -80° C. until further analyses.

(55) 1.6 C-Reactive Protein, Free Hemoglobin and α -Synuclein Quantification

(56) The concentrations of C-reactive protein (CRP) and free hemoglobin were determined in the PFP of all donors using the RayBioTM Human CRP ELISA Kit (RayBiotech, Norcross, GA, USA) and the Hemoglobin Human ELISA kit (Abcam, Toronto, ON, Canada). To quantify α -synuclein (α -Syn) in erythrocytes and EEV, we used the human α -Syn ELISA kit (ThermoFisher Scientific, Waltham, MA, USA). Absorbance values were measured at 450 nm using a multi-detection microplate reader (Synergy HT; BioTek; Winooski, VT, USA). All ELISA tests were performed according to the manufacturer's instructions.

(57) 1.7 Scanning Electron Microscopy

(58) Preparations of erythrocytes (5 μ L) were fixed in 2% paraformaldehyde and 2.5% glutaraldehyde in PBS buffer at least 24 hours before standard dehydration. Samples were washed 3 times for 10 minutes with sodium cacodylate buffer (0.1 M, pH 7.3) and fixed with 1% osmium tetroxide in sodium cacodylate buffer for 90 minutes. Subsequently, samples were washed and processed in 50%, 70%, 90% and 100% ethanol for dehydration (10 minutes/step). Finally, samples were soaked in two subsequent baths of 100% ethanol, for 40 minutes and 10 minutes, air-dried overnight and coated with palladium. Observations were completed using a JEOL 6360LV scanning electron microscope (JEOL, Peabody, MA, USA).

(59) 1.8 Transmission Electron Microscopy

(60) Preparations of EEV (30 μ L) and activated erythrocytes (5 μ L) were fixed in 2% paraformaldehyde at least 24 hours before being dehydrated and sealed in LR white resin. Slices of LR white resin were placed on a Formvar/carbon-coated grid and processed for immunolabeling. The tissues mounted on grids were blocked in 0.5% BSA-c (Aurion, Wageningen, The Netherlands) in HBSS and incubated for 120 minutes with rabbit anti- α -Syn antibody (Abcam, Toronto, ON, Canada) or rabbit anti- α -Syn (phospho S129) antibody (Abcam, Toronto, ON, Canada), both diluted at 1:250 in HBSS and washed several times with distilled water. Finally, the grids were incubated for 60 minutes with an anti-rabbit IgG conjugated to 6 nm gold particles (EMS, Hatfield, PA, USA) diluted at 1:200 and washed several times with distilled water to ultimately be fixed in 2.5% glutaraldehyde (EMS, Hatfield, PA, USA) in HBSS for 15 minutes. For this last step, the grids were treated with 3% uranyl acetate-0.075 M oxalate (pH 7.0) (EMS, Hatfield, PA, USA) for 1 minute, which was followed by several washes in distilled water. Observations were completed with a TECNAI Spirit G2 transmission electron microscope at 80 kV (FEI, Hillsboro, OR, USA).

(61) 1.9 Mass Spectrometry Analysis and Label Free Protein Quantification

(62) For proteomic analyses, EEV from 4 individuals per group (Control, mild PD and moderate PD) were prepared as described above. For each individual, 25 μ g of protein sample, according to Bradford protein assay, were migrated onto an electrophoresis gel 4-12% Bis-Tris to separate hemoglobin from higher proteins. Following gel staining using Sypro Ruby (Thermo Fischer Scientific), the 12 kDa band corresponding to the hemoglobin size was cut out and the remaining part of the gel further fractionated into 7 slices, exposed to trypsin digestion and peptide extraction on a MassPrepTM liquid handling robot (Waters, Milford, USA) according to the manufacturer's specifications and to the protocol of Shevchenko et al., 1996, with the modifications suggested by Havlis et al., 2003. The extracted peptides from the 7 slices of the same individual were pooled and analyzed by nanoLC-MS/MS. The excised hemoglobin gel slices were also analyzed in the same conditions. One μ g of each individual sample was injected on a Dionex UltiMateTM 3000 nanoRSLC system (Thermo Scientific) equipped with a nanoviper Acclaim Pepmap100TM, C18, 3 μ m, 75 μ m $\times$ column (Thermo Scientific) connected to the nanoelectrospray source of an Orbitrap FusionTM mass spectrometer (Thermo Scientific). The peptides were eluted at 300 nL/min using an acetonitrile gradient of 90 minutes and the mass spectrometer was operating in Data Dependent Acquisition mode. Peptide masses were measured in MS spectra detected in the orbitrap at 120K resolution. MSMS fragmentation spectra of peptides were generated by Higher energy Collisional Dissociation (HCD) and detected in the ion trap. Spectra were searched against a human protein database (Uniprot Complete Proteome, taxonomy *Homo sapiens*—83512 sequences) using AndromedaTM search engine included in MaxQuantTM software version 1.5.5.1 (Cox et al., 2008). MaxQuantTM was also used to validate proteins and peptides at 1% False Discovery Rate using a target/decoy database search and to perform Label Free Quantification of the identified proteins using the 'match between runs' option.

(63) 1.10 Further Statistical Analyses

(64) For FIGS. 3C and 3D, data were first tested for normality using the D'Agostino & Pearson

normality test. Comparisons between groups were obtained by Mann-Whitney U test or Kruskal-Wallis ANOVA and performed using Prism 6.0 (GraphPad Software, LaJolla, CA). For analyses displayed in FIG. 4C, the 'Intensity values' contained in the output 'proteingroup.txt' file of MaxQuant™ were used to quantify each identified protein in each individual sample. The values were normalized by the median of each column (all intensity values of proteins for one sample). The missing values were imputed with a noise value corresponding to the 1-percentile of each sample column. For each comparison between two groups (Control, mild PD or moderate PD), proteins with too many imputed values were considered not quantifiable (a minimum of three not-imputed values in one of the 2 groups are required). A protein ratio was calculated between the two groups using the average of intensity values in each group. Finally, a statistical Welch's test was performed between the two groups. The protein ratios were transformed into $\log_2(\text{ratio})$ then centered by calculation of a z-score ($z\text{-score} = (x - \mu) / \sigma$). A protein was considered as variant if it fulfilled the following criteria: minimum of 2 peptides quantified, Welch's test p value < 0.05 and absolute value of z-score > 1.96 (corresponding to values outside of the 95% confidence interval). The Gene Ontology enrichment analysis on the identified proteins (FIG. 4D) was performed on the Cytoscape™ platform (v. 3.4.0) using the BinGO™ software version 3.0.3 (Maere et al., 20005) against all human genes with GO annotation (Uniprot-GOA generated 2015-06-22). Enrichment was calculated by hypergeometric test and Bonferroni Family-Wise Error Rate (FWER) was used to correct for multiple testing. The data for the resulting 8 proteins was standardized, hierarchically clustered and visualized as a heatmap by using the statistical framework R (R Core Team, 2016). The robustness of the nodes was evaluated by computing Approximately Unbiased (AU) p values using the R package pvclust (10000 bootstraps, average method and correlation-based dissimilarity matrix) (Suzuki et al., 2006).

Example 2—Results

(65) The cohorts studied here included Parkinson's disease (PD) (n=60) and Huntington's disease (HD) patients (n=63) of all stages (see Example 1.1), as well as their respective age- and sex-matched healthy controls (n=37; n=55, respectively). The demographics for both cohorts are shown in Table 1. Full blood counts (erythrocytes, lymphocytes, platelets, leukocytes, monocytes, neutrophils) and C-reactive protein (indicative of an inflammatory response) quantification were obtained for all participants, but they did not reveal any significant differences between groups (data not shown). Similarly, the hematocrit, the mean corpuscular hemoglobin, as well as the mean corpuscular volume values were similar between PD and control groups (data not shown).

(66) 2.1 PD Patients Exhibit a Disease-Specific Increase in Erythrocyte-Derived EV

(67) Platelet-free plasma (PFP) and extracellular vesicles (EV) were labeled and quantified according to their cell of origin for all participants, as described in Examples 1.2 and 1.3. Results are summarized in Table 2A (PD patients and controls) and Table 2B (HD patients and controls).

(68) As shown in Table 2A, no significant differences between PD patient and control samples were observed in the number of EV originating from platelets, endothelial cells, monocytes, granulocytes, and leukocytes. Similarly, as shown in Table 2B, no significant differences between HD patient and control samples were observed in the concentrations of EV originating from these same cell types.

(69) Interestingly, a significant increase in erythrocyte-derived EV in patients with PD was observed, as compared to the control group (see values highlighted in black in Table 2A). This increase in erythrocyte-derived EV in patients with PD was disease-specific, as the same effect was not observed in erythrocyte-derived EV in patients with HD (Table 2B).

(70) TABLE-US-00004 TABLE 2A Quantification of extracellular vesicles (EV) derived from different cell types of PD patients and controls

CTRL	PD	P	Cell type	Markers	Units	n	Mean	SEM	n	Mean	SEM
			Platelets	CD41+PS-	× 10 ⁵ /μL	37	7.88	1.68	59	10.3	1.33
				CD41+PS+		37	15.2	3.20	59	17.9	2.53
				CD41+CD31+		37	1.51	0.69	59	1.99	0.54
				CD41+ total		37	23.1	4.62	59	28.2	3.66
			EV	CD41+/platelet		35	0.106	0.021	57	0.125	0.016
				Endothelial cells							

CD31+CD41-PS- ×10.sup.3/μL 37 15.8 8.04 59 11.7 6.37 0.75 CD31+CD41-PS+ 37 0.91 0.13 59 0.92 0.10 0.96 CD31+CD41- total 37 16.7 8.03 59 12.6 6.36 0.75 Monocytes CD45-CD14+ PS- ×10.sup.3/μL 37 1.70 0.30 59 1.62 0.24 0.85 CD45-CD14+ PS+ 37 1.20 4.00 59 5.84 3.17 0.50 CD45+CD14+ PS- 37 0.16 0.04 59 0.14 0.03 0.74 CD45+CD14+PS+ 37 0.60 0.79 59 1.47 0.63 0.59 CD14+ total 37 3.66 4.88 59 9.06 3.87 0.60 EV CD14+/monocyte 35 7.08 1.99 57 9.16 1.56 0.41 Granulocytes CD45-CD15+ PS- ×10.sup.3/μL 37 12.3 7.96 59 16.7 6.30 0.92 CD45-CD15+ PS+ 37 2.21 0.77 59 1.39 0.61 0.47 CD45+CD15+ PS- 37 0.55 0.36 59 1.15 0.29 0.20 CD45+CD15+PS+ 37 1.01 0.30 59 1.25 0.24 0.56 CD15+ total 37 16.0 8.83 59 20.6 6.99 0.91 EV CD15+/granulocyte 35 3.70 0.64 57 3.16 0.50 0.53 Leukocytes CD45+ total ×10.sup.3/μL 37 10.4 2.21 59 13.8 1.75 0.26 Erythrocytes CD235a+PS- ×10.sup.3/μL 36 18.2 46.5 59 32.0 36.3 0.04 CD235a+PS+ 36 0.22 0.07 59 0.29 0.05 0.70 CD235a+ total 36 18.4 47.0 59 32.3 36.7 0.04 EV CD235a+/erythrocyte 34 0.0039 0.011 57 0.0069 0.008 0.04 Abbreviations: CD235a, glycophorin A; EV, extracellular vesicle; PD, Parkinson's disease; PS, phosphatidylserine.

(71) TABLE-US-00005 TABLE 2B Quantification of extracellular vesicles (EV) derived from different cell types of HD patients and controls CTRL HD pre-manifest HD P Cell type Markers Units n Mean SEM n Mean SEM n Mean SEM value Platelets CD41+PS- ×10.sup.3/μL 54 9.2 2.2 10 4.3 1.3 50 6.1 1.0 0.78 CD41+PS+ 54 19.3 4.8 10 7.1 2.0 50 12.4 2.4 0.74 CD41+ total 54 28.4 6.9 10 11.4 3.2 50 18.6 3.4 0.70 EV CD41+/platelet 53 0.12 0.03 10 0.05 0.02 48 0.08 0.01 0.34 Endothelial CD31+CD41-PS- ×10.sup.3/μL 54 1.4 0.3 10 0.6 0.2 50 1.2 0.2 0.31 cells CD31+CD41-PS+ 54 0.68 0.16 10 0.25 0.06 50 0.46 0.09 0.59 CD31+CD41- total 54 2.1 0.4 10 0.8 0.2 50 1.7 0.3 0.26 Monocytes CD45-CD14+ PS- ×10.sup.3/μL 54 3.4 1.1 10 1.6 0.2 51 1.6 0.1 0.91 CD45-CD14+ PS+ 54 1.8 0.3 10 0.8 0.3 51 1.5 0.2 0.14 CD45+CD14+ PS- 54 0.18 0.07 10 0.069 0.016 51 0.056 0.008 0.34 CD45+CD14+PS+ 54 0.62 0.12 10 0.24 0.06 51 0.55 0.14 0.12 CD14+ total 54 6.0 1.3 10 2.6 0.4 51 3.7 0.4 0.08 EV CD14+/monocyte 53 12.3 2.5 10 5.7 0.6 48 8.0 1.0 0.13 Granulocytes CD45-CD15+ PS- ×10.sup.3/μL 54 1.2 0.1 10 1.2 0.3 51 1.5 0.2 0.33 CD45-CD15+ PS+ 54 0.12 0.04 10 0.18 0.08 51 0.22 0.11 0.33 CD45+CD15+ PS- 54 0.20 0.05 10 0.07 0.02 51 0.15 0.04 0.64 CD45+CD15+PS+ 54 0.25 0.05 10 0.13 0.06 51 0.20 0.04 0.39 CD15+ total 54 1.7 0.2 10 1.6 0.4 51 0.20 0.3 0.67 EV CD15+/ 53 0.41 0.04 10 0.42 0.13 48 0.50 0.08 0.75 granulocyte Leukocytes CD45+ total ×10.sup.3/μL 54 33.4 2.7 10 31.6 5.3 51 31.7 2.4 0.88 Erythrocytes CD235a+PS- ×10.sup.3/μL 54 15.2 2.0 10 10.3 3.5 51 14.1 1.4 0.16 CD235a+PS+ 54 1.1 0.2 10 0.4 0.2 51 1.1 0.1 0.04 CD235a+ total 54 16.4 2.0 10 10.7 3.5 51 15.3 1.5 0.09 EV 54 0.0035 0.0005 10 0.0023 0.0008 50 0.0033 0.0003 0.11 CD235a+/erythrocyte Abbreviations: CD235a, glycophorin A; EV, extracellular vesicle; HD, Huntington's disease; PS, phosphatidylserine.

2.2 Increase in Erythrocyte-Derived EV in PD Patient Samples Correlates with PD Progression and PD Treatment

(72) To evaluate its suitability as a potential biomarker for monitoring PD progression, we examined correlations between the number of erythrocyte-derived EV (EEV) and the Unified Parkinson's Disease Rating Scale (UPDRS) PD staging system, because of its greater sensitivity and the recent publications validating this approach (Martinez-Martin et al., 2015). Strikingly, statistical linear regression analysis revealed strong correlations between the number of erythrocyte-derived EV and PD stage/progression. As shown in FIG. 2A, strong correlations (correlations exceeding were observed between the number of erythrocyte-derived EV (expressed as CD135a+ EV/total number of erythrocytes) and patient UPDRS score, and thus PD stages. The R.sup.2 values obtained demonstrated that in both “mild” and “moderate” PD patient groups, at least 87% of the variation in the total number of EEV/erythrocytes is due to the variation of the UPDRS. Moreover, the results are significant with respect to the p values obtained for each fit, since they fall below the 5% confidence level. Hence, the statistical tests on EEV counts uncovered a clear cut-off point between mild and moderate PD patients, which could not be accounted for by daily levodopa doses administered to the patient (FIG. 2A, see details of the 5 patients pinpointed).

(73) The above correlations observed with respect to the number of EEV in PD patients was found to be disease-specific, since a similar analysis performed in HD patients failed to reveal the same strong correlations (see FIG. 2B). In the HD cohort, the total number of EEV between pre-manifest and manifest HD was similar to their age- and sex-matched healthy Controls. In contrast to PD, correlation analyses failed to reveal an association between the number of EEV and HD stage using the United Huntington's Disease Rating Scale (UHDRS) score (FIG. 2B). Based on this, all our subsequent analyses focused only on the PD cohort. It should be re-emphasized that total blood counts did not indicate any significant differences in the number of endothelial cell-, platelet-, monocyte- and granulocyte-derived EV in the platelet-free plasma (PFP) in PD (Table 2A) and HD patients (Table 2B), when compared to their respective control cohorts.

(74) We have thus identified at least two distinct groups of PD patients with highly significant correlations to the number of EV derived from erythrocytes, which relates to PD stage and/or PD treatment (FIG. 2A). Strikingly, these correlations appear to be specific to PD, as similar correlations were not observed in the cohort of HD patients (of varying degrees of severity) in which we performed identical analyses (FIG. 2B).

(75) 2.3 α -Syn is not Differentially Expressed in Normal Vs. Diseased PD Conditions

(76) Having established that EEV counts correlate with disease state in PD, we aimed to assess whether α -Syn—which is not only the main component of Lewy bodies but is highly expressed in most blood cells—was differentially expressed in normal vs. diseased conditions. For this, we opted to use scanning electron microscopy, but this did not reveal any morphological changes between resting and activated erythrocytes in either condition (FIG. 3A). We further used transmission electron microscopy to quantify the number of EEV containing α -Syn and phosphorylated (serine 129) forms of the protein but again no significant differences between PD patients and age- and sex-matched healthy Controls (FIGS. 3B and 3C) were observed. Quantified α -Syn levels in EEV from PD patients and Controls using commercial ELISA kits corroborated these results (FIG. 3D).

(77) 2.4 Analysis of the EEV Proteome Revealed 8 Differentially Expressed Proteins

(78) Since our combined quantifications (ELISA and transmission electron microscopy) suggested that α -Syn levels could not be used as a blood marker of disease, we sought to obtain the specific protein signature of EEV from mild and moderate PD patients (with respect to the UPDRS scores) and their age-matched Controls. Given the significant amounts of hemoglobin within erythrocytes that could mask the true nature of the protein signature in EEV, we performed a label free quantitative proteomic analysis by nanoLC/MSMS (Wither et al., 2016) using two distinct approaches: with and without hemoglobin.

(79) By removing the hemoglobin, we identified a total of 818 proteins in comparison with 356 when we did not perform this methodological step (refer to Table 4 for complete list of proteins), which clearly, provides a much more thorough evaluation of the protein content of EEV. Additionally, a Gene Ontology enrichment analysis on the 'Cellular Component' ontology performed on the two sets of identified proteins in comparison with the whole human proteome, revealed that our samples are enriched with elements associated to 'vesicles' and 'hemoglobin complex' which show the efficiency of our EEV production and purification protocol (data not shown).

(80) Out of the 818 proteins identified in the proteome of EEV, 8 had their expression significantly modified according to the different stages of PD (FIGS. 4A and 4B). Hierarchical clustering, coupled to a heatmap (FIG. 4D), allowed us to group individuals according to stages of disease (Control, mild PD and moderate PD) and provided compelling evidence that the 8 proteins identified could also be grouped into three categories. Proteins of group I were highly and predominantly expressed in Controls, proteins of group II were highly and predominantly expressed in mild PD patients, and proteins belonging to group III were highly and predominantly expressed in moderate PD patients (FIGS. 4B and 4C; Table 3). This data set was further confirmed

by volcano plots (FIG. 5). Of note, two proteins of group I are associated with the regulation system of the cell (ABHD148, NADSYN1) and one protein significantly expressed in moderate PD patients (ATP5A1) is involved in the regulation of mitochondrial ATP.

(81) TABLE-US-00006 TABLE 3 EEV proteins differentially expressed according to different stages of PD Groups Protein Gene I Axin interactor, dorsalization-associated AIDA (Control) protein Alpha/beta hydrolase domain-containing ABHD14B protein 14B Glutamine-dependent NAD(+) synthetase NADSYN1 II Dihydropteridine reductase QDPR (mild PD) Alcohol dehydrogenase [NADP(+)] AKR1A1 CB1 cannabinoid receptor-interacting CNRIP1 protein 1 III Ubiquitin carboxyl-terminal hydrolase USP24 (moderate 24 PD) ATP synthase subunit alpha, ATP5A1 mitochondrial

(82) Fold changes and the results of statistical analyses of the proteins of Table 3 are shown in Table 3.1, in which the proteins are listed in order of their “fold” difference for each group comparison. For example, a “fold” of 1.931 indicates that the AKR1A1 protein was detected in the isolated EEV preparations almost two times higher in the mild group, as compared to the control group.

(83) TABLE-US-00007 TABLE 3.1 Fold changes and statistics of EEV proteins of Table 3 Protein Gene Welch's test Fold change Z-score Mild (Group II) v. Controls (Group I) Alcohol dehydrogenase [NADP(+)] AKR1A1 0.019 1.931 2.25 Alpha/beta hydrolase domain-containing protein 14B ABHD14B 0.016 0.274 -3.57 Moderate (Group III) v. Controls (Group I) ATP synthase subunit alpha, mitochondrial ATP5A1 0.028 3.906 3.25 Glutamine-dependent NAD(+) synthetase NADSYN1 0.009 0.391 -2.31 Axin interactor, dorsalization-associated protein AIDA 0.046 0.379 -2.38 Alpha/beta hydrolase domain-containing protein 14B ABHD14B 0.017 0.282 -3.09 Moderate (Group III) v. Mild (Group II) ATP synthase subunit alpha, mitochondrial ATP5A1 0.027 3.629 2.77 Ubiquitin carboxyl-terminal hydrolase 24 USP24 0.036 2.971 2.29 CB1 cannabinoid receptor-interacting protein 1 CNRIP1 0.014 0.352 -2.80 Alcohol dehydrogenase [NADP(+)] AKR1A1 0.002 0.342 -2.87 Dihydropteridine reductase QDPR 0.040 0.341 -2.88 Proteins were considered as significantly differentially expressed if they respected two conditions: Z-score >1.96 or <-1.96, and Welch's test p-value < 0.05.

(84) As seen in Table 3.1 and in FIG. 4C, the ABHD14B protein (alpha/beta hydrolase domain-containing protein 14B) was detected over 3.5-fold lower both in the mild v. controls and in the moderate v. controls groups, suggesting that this EEV protein may be a useful biomarker for clinically assessing symptomatic PD subjects. The results in Table 3.1 and in FIG. 4C suggest that the AKR1A1 protein (alcohol dehydrogenase [NADP(+)]) may be a useful biomarker for clinically assessing mild PD subjects, as this EEV protein was detected almost 2-fold higher in mild v. control PD subjects, but was not differentially expressed in moderate v. control PD subjects. Finally, the results in Table 3.1 and in FIG. 4C suggest that the ATP5A1 protein may be a useful biomarker for clinically assessing moderate PD subjects, since this protein was detected over 3.6-fold higher in both moderate v. controls and in moderate v. mild PD subjects.

Example 3—Discussion

(85) We have identified biomarkers correlating to different states of PD based on the quantification of EV shed from erythrocytes and UPDRS scores. In particular, we have shown a strong correlation between the number of EEV and the clinical expression/stages of PD. Finding this level of correlation in a 60-patient sample size in such a heterogeneous disorder highlights the robustness of the biomarkers identified herein.

(86) Although staging of PD is often done using the H & Y clinical scale, we sought to use the UPDRS given its greater sensitivity and the recent publications validating this approach (Martinez-Martin et al., 2015). Using these scores, we found that mild PD patients—with a UPDRS score lower than 37—are characterized by an increased number of EV during the mild disease stage (correlations=0.886); and that the same pattern repeated itself with patients who had UPDRS scores between 37 and 75 (correlations=0.873). When comparing individual patients from both correlation

curves, the dose of levodopa taken daily did not seem to account for the differences, as there was no correlation between the levodopa dose and the number of EEV.

(87) Finally, we herein report an improved method to perform proteomic analysis of EEV in blood samples by removing hemoglobin, a large protein that can easily mask other ones within a protein signature. Indeed, the high dynamic range of protein concentrations in erythrocytes and therefore in EEV, due to the high abundance of hemoglobin, decreases the capacity of the mass spectrometer to detect signals corresponding to low abundance proteins. Analyzing the hemoglobin separately from the other proteins of other molecular weights allowed us to go deeper in the EEV proteome by identifying 129% more proteins than in the initial analysis.

(88) TABLE-US-00008 TABLE 4 Proteins identified in the EEV proteome. Lists of the proteins identified by LC-MS/MS analysis using Andromeda™/MaxQuant™ search engine in the Uniprot Complete Proteome *Homo sapiens* database for the non-hemoglobin-depleted sample (356 proteins) (A), or when the hemoglobin-depleted and hemoglobin-containing fractions were analyzed separately (818 proteins) (B). The lists were filtered at 1% False Discovery Rate using a target/decoy database search. Razor + unique Protein ID Protein names Gene names peptides (A). UNIPROT COMPLETE PROTEOME *HOMO SAPIENS* DATABASE FOR THE NON-DEPLETED SAMPLE P02549 Spectrin alpha chain, erythrocytic 1 SPTA1 136 P11277 Spectrin beta chain, erythrocytic SPTB 123 P16157 Ankyrin-1 ANK1 70 P02730 Band 3 anion transport protein SLC4A1 38 P55072 Transitional endoplasmic reticulum ATPase VCP 33 P16452 Erythrocyte membrane protein band 4.2 EPB42 33 P11171 Protein 4.1 EPB41 33 P04040 Catalase CAT 32 Q13228 Selenium-binding protein 1 SELENBP1 29 B4DT77 Annexin; Annexin A7 ANXA7 27 Q8WUM4 Programmed cell death 6-interacting protein PDCD6IP 27 P35612 Beta-adducin ADD2 26 P68871 Hemoglobin subunit beta; LW-hemorphin-7; Spinorphin HBB 25 Q00610 Clathrin heavy chain 1 CLTC 25 P69905 Hemoglobin subunit alpha HBA1 22 B4DVE7 Annexin A11 ANXA11 20 J3QLD9 Flotillin-2 FLOT2 20 P08758 Annexin A5; Annexin ANXA5 19 P09525 Annexin A4; Annexin ANXA4 18 O75955; Flotillin-1 FLOT1 18 P11142 Heat shock cognate 71 kDa protein HSPA8 18 P32119 Peroxiredoxin-2 PRDX2 17 P00491 Purine nucleoside phosphorylase PNP 17 P27105 Erythrocyte band 7 integral membrane protein STOM 17 P00918 Carbonic anhydrase 2 CA2 17 P23634 Plasma membrane calcium-transporting ATPase 4 ATP2B4 17 C9JIF9 Acylamino-acid-releasing enzyme APEH 17 P00915 Carbonic anhydrase 1 CA1 16 P63261 Actin, cytoplasmic 2 ACTG1 16 Q5VU58 Tropomyosin alpha-3 chain TPM3 16 P30041 Peroxiredoxin-6 PRDX6 16 E7EU23 Rab GDP dissociation inhibitor beta GDI2 16 Q00013 55 kDa erythrocyte membrane protein MPP1 16 E7EV01 Calpain-5 CAPN5 16 J3KPS3 Fructose-bisphosphate aldolase A ALDOA 15 Q08495 Dematin DMTN 15 P23276 Kell blood group glycoprotein KEL 15 P69892 Hemoglobin subunit gamma-2 HBG2 14 P60174 Triosephosphate isomerase TPI1 14 P62258 14-3-3 protein epsilon YWHAE 14 O75326 Semaphorin-7A SEMA7A 14 O75340 Programmed cell death protein 6 PDCD6 13 E7EV99 Alpha-adducin ADD1 13 Q5VZU9 Tripeptidyl-peptidase 2 TPP2 13 P53396 ATP-citrate synthase ACLY 13 C9J0K6 Sorcin SRI 12 P04406 Glyceraldehyde-3-phosphate dehydrogenase GAPDH 12 P07738 Bisphosphoglycerate mutase BPGM 12 B7Z3I9 Delta-aminolevulinic acid dehydratase ALAD 12 Q86X55 Histone-arginine methyltransferase CARM1 CARM1 12 P07195 L-lactate dehydrogenase B chain; L-lactate dehydrogenase LDHB 12 P23526 Adenosylhomocysteinase AHCY 12 Q32Q12 Nucleoside diphosphate kinase NME1-NME2 12 B7Z7A9 Phosphoglycerate kinase 1 PGK1 11 P11166 Solute carrier family 2, facilitated glucose transporter member 1 SLC2A1 11 P22303 Acetylcholinesterase ACHE 11 Q9NP58 ATP-binding cassette sub-family B member 6, mitochondrial ABCB6 11 P40925 Malate dehydrogenase, cytoplasmic; Malate dehydrogenase MDH1 11 P00352 Retinal dehydrogenase 1 ALDH1A1 11 F2Z2V0 Copine-1 CPNE1 11 K7EMC9 WW domain-binding protein 2 WBP2 10 F5H7S3 Tropomyosin alpha-1 chain TPM1 10 A6NN80 Annexin A6; Annexin ANXA6 10 P30043 Flavin reductase (NADPH) BLVRB 9 H7BXD5 Grancalcin GCA 9 P04083 Annexin A1; Annexin ANXA1 9 P62937 Peptidyl-prolyl cis-trans

isomerase PPIA 9 P37837 Transaldolase TALDO1 9 Q06830 Peroxiredoxin-1 PRDX1 9 O75131
 Copine-3 CPNE3 9 P00390 Glutathione reductase, mitochondrial GSR 9 E7EQB2 Lactotransferrin
 LTF 9 P07384 Calpain-1 catalytic subunit CAPN1 9 P02042 Hemoglobin subunit delta HBD 8
 P30086 Phosphatidylethanolamine-binding protein 1; Hippocampal cholinergic neurostimulating
 PEBP1 8 peptide P35613 Basigin BSG 8 Q9H0U4 Ras-related protein Rab-1B; Putative Ras-
 related protein Rab-1C RAB1B; RAB1C 8 P63092 Guanine nucleotide-binding protein G(s)
 subunit alpha isoforms GNAS 8 P48506 Glutamate-cysteine ligase catalytic subunit GCLC 8
 P06702 Protein S100-A9 S100A9 7 Q9UBV8 Peflin PEF1 7 P17931 Galectin-3; Galectin
 LGALS3 7 P28066 Proteasome subunit alpha type-5 PSMA5 7 P07451 Carbonic anhydrase 3 CA3
 7 E7EQ12 Calpastatin CAST 7 P50895 Basal cell adhesion molecule BCAM 7 P28074 Proteasome
 subunit beta type-5 PSMB5 7 G3V5Z7 Proteasome subunit alpha type; Proteasome subunit alpha
 type-6 PSMA6 7 P25786 Proteasome subunit alpha type-1 PSMA1 7 G3V1D3 Dipeptidyl
 peptidase 3 DPP3 7 P49247 Ribose-5-phosphate isomerase RPIA 7 Q5T9B7 Adenylate kinase
 isoenzyme 1 AK1 7 P25789 Proteasome subunit alpha type-4; Proteasome subunit beta type
 PSMA4 7 B4E022 Transketolase TKT 7 J3QS39 Ubiquitin-60S ribosomal protein L40; Ubiquitin;
 60S ribosomal protein L40; Ubiquitin-40S UBB; RPS27A; 6 ribosomal protein S27a; Ubiquitin;
 40S ribosomal protein S27a; Polyubiquitin- UBC; UBA52; B; Ubiquitin; Polyubiquitin-C;
 Ubiquitin UBBP4 H0Y7A7 Calmodulin CALM2 6 P28070 Proteasome subunit beta type-4
 PSMB4 6 Q9H4G4 Golgi-associated plant pathogenesis-related protein 1 GLIPR2 6 Q9BY43
 Charged multivesicular body protein 4a CHMP4A 6 P48426 Phosphatidylinositol 5-phosphate 4-
 kinase type-2 alpha PIP4K2A 6 P28289 Tropomodulin-1 TMOD1 6 P07911 Uromodulin;
 Uromodulin, secreted form UMOD 6 Q9GZP4 PITH domain-containing protein 1 PITHD1 6
 P78417 Glutathione S-transferase omega-1 GSTO1 6 P25788 Proteasome subunit alpha type-3
 PSMA3 6 O14818 Proteasome subunit alpha type-7; Proteasome subunit alpha type-7-like
 PSMA7; PSMA8 6 P08107 Heat shock 70 kDa protein 1A/1B HSPA1A 6 H0YD13 CD44 antigen
 CD44 6 P61225 Ras-related protein Rap-2b; Ras-related protein Rap-2c; Ras-related protein Rap-
 2a RAP2B; RAP2A; 6 RAP2C P05109 Protein S100-A8; Protein S100-A8, N-terminally processed
 S100A8 6 P23528 Cofilin-1 CFL1 6 Q99808 Equilibrative nucleoside transporter 1 SLC29A1 6
 P84077 ADP-ribosylation factor 1; ADP-ribosylation factor 3; ADP-ribosylation factor 5; ADP-
 ARF1; ARF3; 6 ribosylation factor 4 ARF5; ARF4 P31946 14-3-3 protein beta/alpha; 14-3-3
 protein beta/alpha, N-terminally processed YWHAB 6 C9JIS1 Guanine nucleotide-binding protein
 G(I)/G(S)/G(T) subunit beta-2; Guanine nucleotide- GNB2; GNB4 6 binding protein subunit beta-4
 P53990 IST1 homolog IST1 6 Q99497 Protein DJ-1 PARK7 6 F5H7U0 6-phosphogluconate
 dehydrogenase, decarboxylating PGD 6 B7Z7E9 Aspartate aminotransferase, cytoplasmic GOT1 6
 P62834 Ras-related protein Rap-1A; Ras-related protein Rap-1b; Ras-related protein Rap-1b-like
 RAP1A; RAP1B 6 protein P04899 Guanine nucleotide-binding protein G(i) subunit alpha-2
 GNAI2 6 P25325 3-mercaptopyruvate sulfurtransferase; Sulfurtransferase MPST 6 Q9NP79
 Vacuolar protein sorting-associated protein VTA1 homolog VTA1 6 P00492 Hypoxanthine-guanine
 phosphoribosyltransferase HPRT1 6 Q16531 DNA damage-binding protein 1 DDB1 6 P22314
 Ubiquitin-like modifier-activating enzyme 1 UBA1 6 P00441 Superoxide dismutase [Cu—Zn]
 SOD1 5 H7BY58 Protein-L-isoaspartate O-methyltransferase; Protein-L-isoaspartate(D-aspartate)
 O- PCMT1 5 methyltransferase P10768 S-formylglutathione hydrolase ESD 5 P09543 2,3-cyclic-
 nucleotide 3-phosphodiesterase CNP 5 P06733 Alpha-enolase; Enolase ENO1 5 P26038 Moesin
 MSN 5 O75368 SH3 domain-binding glutamic acid-rich-like protein SH3BGRL 5 K7EQ48
 Glucose-6-phosphate isomerase GPI 5 P25787 Proteasome subunit alpha type-2 PSMA2 5 P49721
 Proteasome subunit beta type-2 PSMB2 5 F5H8J2 Protein disulfide-isomerase P4HB 5 C9J9P4
 Phospholipid scramblase 1 PLSCR1 5 P09211 Glutathione S-transferase P GSTP1 5 B5MDF5
 GTP-binding nuclear protein Ran RAN 5 P07900 Heat shock protein HSP 90-alpha HSP90AA1 5
 K7EQ02 DAZ-associated protein 1 DAZAP1 5 Q13630 GDP-L-fucose synthase TSTA3 5 F5H0T1
 Stress-induced-phosphoprotein 1 STIP1 5 P50502 Hsc70-interacting protein; Putative protein

FAM10A5; Putative protein FAM10A4 ST13; ST13P5; 5 ST13P4 P20618 Proteasome subunit beta type-1 PSMB1 5 P62805 Histone H4 HIST1H4A 5 P51148 Ras-related protein Rab-5C RAB5C 5 H7C2G2 Ecto-ADP-ribosyltransferase 4 ART4 5 J3KQ18 D-dopachrome decarboxylase; D-dopachrome decarboxylase-like protein DDT; DDTL 5 H3BPK3 Hydroxyacylglutathione hydrolase, mitochondrial HAGH 5 B4DIT7 Protein-glutamine gamma-glutamyltransferase 2 TGM2 5 O43633 Charged multivesicular body protein 2a CHMP2A 5 B4DQH4 T-complex protein 1 subunit theta CCT8 5 Q9UN37 Vacuolar protein sorting-associated protein 4A VPS4A 5 I3L397 Eukaryotic translation initiation factor 5A-1; Eukaryotic translation initiation factor 5A-1-like EIF5A; EIF5AL1 5 Q9Y5Z4 Heme-binding protein 2 HEBP2 5 Q9UKV8 Protein argonaute-2 AGO2 5 F5H442 Tumor susceptibility gene 101 protein TSG101 5 H3BLV0 Complement decay-accelerating factor CD55 5 P02008 Hemoglobin subunit zeta HBZ 5 E7EPV7 Alpha-synuclein SNCA 4 P63104 14-3-3 protein zeta/delta YWHAZ 4 P02724 Glycophorin-A GYPA; GP_{Erik} 4 U3KQE2 Calpain small subunit 1 CAPNS1 4 Q9NP59 Solute carrier family 40 member 1 SLC40A1 4 Q5QPM9 Proteasome inhibitor PI31 subunit PSMF1 4 C9J8U2 Nicotinate phosphoribosyltransferase NAPRT 4 P61981 14-3-3 protein gamma; 14-3-3 protein gamma, N-terminally processed YWHAG 4 Q9UQ80 Proliferation-associated protein 2G4 PA2G4 4 E5RJR5 S-phase kinase-associated protein 1 SKP1 4 Q9NRV9 Heme-binding protein 1 HEBP1 4 Q9H444 Charged multivesicular body protein 4b CHMP4B 4 P10599 Thioredoxin TXN 4 Q5VSJ9 Blood group Rh(CE) polypeptide; Blood group Rh(D) polypeptide RHCE; RHD 4 Q9UK41 Vacuolar protein sorting-associated protein 28 homolog VPS28 4 Q9Y3I1 F-box only protein 7 FBXO7 4 P61026 Ras-related protein Rab-10 RAB10 4 Q14974 Importin subunit beta-1 KPNB1 4 P27797 Calreticulin CALR 4 P18669 Phosphoglycerate mutase 1; Probable phosphoglycerate mutase 4 PGAM1; 4 PGAM4 P54725 UV excision repair protein RAD23 homolog A RAD23A 4 P30613 Pyruvate kinase PKLR PKLR 4 P63000 Ras-related C3 botulinum toxin substrate 1; Ras-related C3 botulinum toxin substrate 3; Ras- RAC1; RAC3; 4 related C3 botulinum toxin substrate 2 RAC2 P05164 Myeloperoxidase MPO 4 F5GWY2 Bifunctional purine biosynthesis protein PURH; Phosphoribosylaminoimidazolecarboxamide ATIC 4 formyltransferase; IMP cyclohydrolase P48637 Glutathione synthetase GSS 4 F5H5V4 26S proteasome non-ATPase regulatory subunit 9 PSMD9 4 G5E9R5 Low molecular weight phosphotyrosine protein phosphatase ACP1 3 P46976 Glycogenin-1 GYG1 3 P28072 Proteasome subunit beta type-6 PSMB6 3 P26447 Protein S100-A4 S100A4 3 F5GXQ0 BRO1 domain-containing protein BROX BROX 3 P08754 Guanine nucleotide-binding protein G(k) subunit alpha GNAI3 3 Q99436 Proteasome subunit beta type-7 PSMB7 3 P62942 Peptidyl-prolyl cis-trans isomerase FKBP1A 3 U3KQK0 Histone H2B HIST1H2B 3 J3QKR3 Proteasome subunit beta type-3 PSMB3 3 P01116 GTPase KRas; GTPase KRas, N-terminally processed; GTPase HRas; GTPase HRas, N- KRAS; HRAS; 3 terminally processed; GTPase NRas NRAS P13489 Ribonuclease inhibitor RNH1 3 Q08722 Leukocyte surface antigen CD47 CD47 3 Q5T123 SH3 domain-binding glutamic acid-rich-like protein 3 SH3BGRL3 3 Q8WYQ7 Galectin; Galectin-9 LGALS9 3 O75695 Protein XRP2 RP2 3 P00167 Cytochrome b5 CYB5A 3 Q9Y4D1 Disheveled-associated activator of morphogenesis 1 DAAM1 3 P11021 78 kDa glucose-regulated protein HSPA5 3 H7C1D4 Translin TSN 3 P07737 Profilin-1 PFN1 3 M0R389 Platelet-activating factor acetylhydrolase IB subunit gamma PAFAH1B3 3 A6NJA2 Ubiquitin carboxyl-terminal hydrolase 14 USP14 3 P10644 cAMP-dependent protein kinase type I-alpha regulatory subunit PRKAR1A 3 Q9BS40 Latexin LXN 3 G5EA52 Protein disulfide-isomerase A3 PDIA3 3 P53004 Biliverdin reductase A BLVRA 3 Q04656 Copper-transporting ATPase 1 ATP7A 3 H9KV70 Neutrophil gelatinase-associated lipocalin LCN2 3 O00299 Chloride intracellular channel protein 1 CLIC1 3 F8WF69 Clathrin light chain A CLTA 3 G3V2F7 Ubiquitin-conjugating enzyme E2 variant 1; Ubiquitin-conjugating enzyme E2 variant 2 UBE2V1; 3 UBE2V2 F8WDS9 LanC-like protein 1 LANCL1 3 P60891 Ribose-phosphate pyrophosphokinase 1 PRPS1 3 K7ESE8 Bleomycin hydrolase BLMH 3 H0YNE3 Proteasome activator complex subunit 1 PSME1 3 P16930 Fumarylacetoacetase FAH 3

F8VSD4 Ubiquitin-conjugating enzyme E2 N UBE2N 3 P07203 Glutathione peroxidase 1 GPX1 3
 P62328 Thymosin beta-4; Hematopoietic system regulatory peptide TMSB4X; 3 TMSB4XP4
 E5RIW3 Tubulin-specific chaperone A TBCA 3 M0R0Y2 Alpha-soluble NSF attachment protein
 NAPA 3 P15374 Ubiquitin carboxyl-terminal hydrolase isozyme L3 UCHL3 3 P04921
 Glycophorin-C GYPC 2 H0YDI1 Lymphocyte function-associated antigen 3 CD58 2 B4E220
 Aquaporin-1 AQP1 2 C9JEN3 Protein lifeguard 3 TMBIM1 2 F5H2R5 Rho GDP-dissociation
 inhibitor 2 ARHGDIB 2 Q53TN4 Cytochrome b reductase 1 CYBRD1 2 Q9NZD4 Alpha-
 hemoglobin-stabilizing protein AHSP 2 Q8NHG7 Small VCP/p97-interacting protein SVIP 2
 Q5JYX0 Cell division control protein 42 homolog CDC42 2 Q71RC9 Small integral membrane
 protein 5 SMIM5 2 E9PNW4 CD59 glycoprotein CD59 2 P09105 Hemoglobin subunit theta-1
 HBQ1 2 R4GN98 Protein S100; Protein S100-A6 S100A6 2 O75531 Barrier-to-autointegration
 factor; Barrier-to-autointegration factor, N-terminally processed BANF1 2 Q5T6W5
 Heterogeneous nuclear ribonucleoprotein K HNRNPK 2 F5H4Q5 Vacuolar protein sorting-
 associated protein 37C VPS37C 2 J3QK90 NSFL1 cofactor p47 NSFL1C 2 H3BV85 Bola-like
 protein 2 BOLA2B; BOLA2 2 Q9NRX4 14 kDa phosphohistidine phosphatase PHPT1 2 H3BS66
 Small integral membrane protein 1 SMIM1 2 E7ESC6 Exportin-7 XPO7 2 P68402 Platelet-
 activating factor acetyl hydrolase IB subunit beta PAFAH1B2 2 Q9BRF8 Serine/threonine-protein
 phosphatase CPPED1 CPPED1 2 P08246 Neutrophil elastase ELANE 2 E9PN50 26S protease
 regulatory subunit 6A PSMC3 2 E7EUC7 UTP-glucose-1-phosphate uridylyltransferase UGP2 2
 B8ZZB8 CB1 cannabinoid receptor-interacting protein 1 CNRIP1 2 E9PCS3 26S proteasome non-
 ATPase regulatory subunit 2 PSMD2 2 P59666 Neutrophil defensin 3; HP 3-56; Neutrophil
 defensin 2; Neutrophil defensin 1; HP 1- DEFA3; DEFA1 2 56; Neutrophil defensin 2 O15400
 Syntaxin-7 STX7 2 P00338 L-lactate dehydrogenase A chain LDHA 2 P61970 Nuclear transport
 factor 2 NUTF2 2 E7EMV0 Protein diaphanous homolog 1 DIAPH1 2 F5GY90 Porphobilinogen
 deaminase HMBS 2 P61020 Ras-related protein Rab-5B RAB5B 2 Q99828 Calcium and integrin-
 binding protein 1 CIB1 2 B4DUA0 Plastin-2 LCP1 2 C9JTY3 Protein TFG TFG 2 P27348 14-3-3
 protein theta YWHAG 2 H0YKZ7 Annexin; Annexin A2; Putative annexin A2-like protein
 ANXA2; 2 ANXA2P2 P08238 Heat shock protein HSP 90-beta HSP90AB1 2 J3KQP6 Ras-related
 protein Rab-11B; Ras-related protein Rab-11A RAB11A; 2 RAB11B A6NMU3 Signal transducing
 adapter molecule 1 STAM 2 P53985 Monocarboxylate transporter 1 SLC16A1 2 F6USW4 F-actin-
 capping protein subunit beta CAPZB 2 O14964 Hepatocyte growth factor-regulated tyrosine kinase
 substrate HGS 2 P20020 Plasma membrane calcium-transporting ATPase 1; Calcium-transporting
 ATPase ATP2B1 2 P36959 GMP reductase 1 GMPR 2 Q9Y376 Calcium-binding protein 39
 CAB39 2 Q9Y6M5 Zinc transporter 1 SLC30A1 2 Q8IZ83 Aldehyde dehydrogenase family 16
 member A1 ALDH16A1 2 Q99459 Cell division cycle 5-like protein CDC5L 2 P06132
 Uroporphyrinogen decarboxylase UROD 2 J3KNT0 Fascin FSCN1 2 P49189 4-
 trimethylaminobutyraldehyde dehydrogenase ALDH9A1 2 H3BNT7 26S proteasome non-ATPase
 regulatory subunit 7 PSMD7 2 P05023 Sodium/potassium-transporting ATPase subunit alpha-1;
 Sodium/potassium-transporting ATP1A1; 2 ATPase subunit alpha-3; Sodium/potassium-
 transporting ATPase subunit alpha- ATP1A2; 2; Sodium/potassium-transporting ATPase subunit
 alpha-4; Potassium-transporting ATPase ATP1A3; alpha chain 1; Potassium-transporting ATPase
 alpha chain 2 ATP1A4; ATP4A; ATP12A P34932 Heat shock 70 kDa protein 4 HSPA4 2 K7EMV3
 Histone H3 H3F3B 2 Q8IU68 Transmembrane channel-like protein 8 TMC8 2 E7ENZ3 T-complex
 protein 1 subunit epsilon CCT5 2 Q5TZA2 Rootletin CROCC 2 Q9P203 BTB/POZ domain-
 containing protein 7 BTBD7 2 Q7LBR1 Charged multivesicular body protein 1b CHMP1B 2
 U3KQ56 Glyoxylate reductase/hydroxypyruvate reductase GRHPR 2 H0YJ11 Alpha-actinin-1;
 Alpha-actinin-2; Alpha-actinin-4 ACTN1; ACTN4; 2 ACTN2 Q9UDT6 CAP-Gly domain-
 containing linker protein 2 CLIP2 2 P09960 Leukotriene A-4 hydrolase LTA4H 2 Q5HY54
 Filamin-A FLNA 2 E9PJL5 Uncharacterized protein C12orf55; Putative uncharacterized protein
 C12orf63 C12orf55; 2 C12orf63 G3V2U7 Acylphosphatase; Acylphosphatase-1 FBN3; ACYP1 2

E9PQN4 Complement receptor type 1 CR1 2 Q9BSL1 Ubiquitin-associated domain-containing protein 1 UBAC1 2 Q04917 14-3-3 protein eta YWHAH 2 B7ZBP9 Serine/threonine-protein phosphatase 2A activator PPP2R4; DKF 2 Zp781M17165 Q8NDC0 MAPK-interacting and spindle-stabilizing protein-like MAPK1IP1L 1 P68133 Actin, alpha skeletal muscle; Actin, alpha cardiac muscle 1; Actin, gamma-enteric smooth ACTA1; ACTC1; 1 muscle; Actin, aortic smooth muscle ACTG2; ACTA2 P69891 Hemoglobin subunit gamma-1 HBG1 1 S4R3Y4 Protein AM BP; Alpha-1-microglobulin; Inter-alpha-trypsin inhibitor light chain; Trypstatin AMBP 1 I3L3E4 Charged multivesicular body protein 6 CHMP6 1 Q16570 Atypical chemokine receptor 1 ACKR1 1 Q5VY30 Retinol-binding protein 4; Plasma retinol-binding protein(1-182); Plasma retinol-binding RBP4 1 protein(1-181); Plasma retinol-binding protein(1-179); Plasma retinol-binding protein(1-176) E7END7 Ras-related protein Rab-1A RAB1A 1 Q5VU59 TPM3 1 P17066 Heat shock 70 kDa protein 6; Putative heat shock 70 kDa protein 7 HSPA6; HSPA7 1 Q04760 Lactoylglutathione lyase GLO1 1 D6RD66 WD repeat-containing protein 1 WDR1 1 K7EM02 Katanin p60 ATPase-containing subunit A-like 2 KATNAL2 1 P14209 CD99 antigen CD99 1 E9PIR7 Thioredoxin reductase 1, cytoplasmic GML; TXNRD1 1 K7EMQ9 EIF3K 1 P15531 Nucleoside diphosphate kinase A NME1 1 H7BZT4 Small ubiquitin-related modifier 4; Small ubiquitin-related modifier 2; Small ubiquitin-related SUMO2; SUMO3; 1 modifier 3 SUMO4 O00560 Syntenin-1 SDCBP 1 Q9BVM4 Gamma-glutamylaminocyclotransferase GGACT 1 K7EKH5 Fructose-bisphosphate aldolase C ALDOC 1 P49773 Histidine triad nucleotide-binding protein 1 HINT1 1 H0YBY6 Disks large-associated protein 2 DLGAP2 1 Q9Y624 Junctional adhesion molecule A F11R 1 B1AKQ8 Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-1; Guanine nucleotide- GNB1; GNB3 1 binding protein G(I)/G(S)/G(T) subunit beta-3 K7EKN6 Urea transporter 1 SLC14A1 1 I3L0K2 Thioredoxin domain-containing protein 17 TXNDC17 1 A8MXY0 Syntaxin-4 STX4 1 O14773 Tripeptidyl-peptidase 1 TPP1 1 E9PNW0 Nucleosome assembly protein 1-like 1; Nucleosome assembly protein 1-like 4 NAP1L4; 1 NAP1L1 Q5TDH0 Protein DDI1 homolog 2 DDI2 1 Q96JM4 Leucine-rich repeat and IQ domain-containing protein 1 LRRIQ1 1 F5GWT9 Phosphoribosylformylglycinamide synthase PFAS 1 F2Z3J2 26S proteasome non-ATPase regulatory subunit 5 PSMD5 1 J3QL74 Zinc finger and BTB domain-containing protein 14 ZBTB14 1 E9PJC7 CD82 antigen CD82 1 Q9H936 Mitochondrial glutamate carrier 1 SLC25A22 1 D6RD63 COP9 signalosome complex subunit 4 COPS4 1 Q6B0K9 Hemoglobin subunit mu HBM 1 Q31611 HLA class I histocompatibility antigen, alpha chain G HLA-G 1 H7BY04 Laminin subunit gamma-3 LAMC3 1 Q9UL25 Ras-related protein Rab-21 RAB21 1 H7C3P7 Ras-related protein Ral-A RALA 1 P08311 Cathepsin G CTSG 1 E9PE37 Ras-related protein Rab-2A; Ras-related protein Rab-2B RAB2B; RAB2A 1 G3V1N2 HBA2 1 P00387 NADH-cytochrome b5 reductase 3; NADH-cytochrome b5 reductase 3 membrane-bound CYB5R3 1 form; NADH-cytochrome b5 reductase 3 soluble form O75339 Cartilage intermediate layer protein 1; Cartilage intermediate layer protein 1 C1; Cartilage CILP 1 intermediate layer protein 1 C2 P14324 Farnesyl pyrophosphate synthase FDPS 1 K7EKG2 Thioredoxin-like protein 1 TXNL1 1 (B). UNIPROT COMPLETE PROTEOME *HOMO SAPIENS* DATABASE WHEN THE HEMOGLOBIN WAS ANALYZED SEPARATELY P02549 Spectrin alpha chain, erythrocytic 1 SPTA1 197 P11277 Spectrin beta chain, erythrocytic SPTB 177 P16157 Ankyrin-1 ANK1 94 P55072 Transitional endoplasmic reticulum ATPase VCP 53 P111714 Protein 4.1 EPB41 52 P35579 Myosin-9 MYH9 44 Q8WUM4 Programmed cell death 6-interacting protein PDCD6IP 43 P02730 Band 3 anion transport protein SLC4A1 43 P16452 Erythrocyte membrane protein band 4.2 EPB42 42 P04040 Catalase CAT 40 A0A087WVQ6 Clathrin heavy chain; Clathrin heavy chain 1 CLTC 38 P35612 Beta-adducin ADD2 37 P16157 Ankyrin-1 ANK1 36 Q14254 Flotillin-2 FLOT2 33 P20073 Annexin A7 ANXA7 31 O75955 Flotillin-1 FLOT1 30 P53396 ATP-citrate synthase ACLY 30 P06753 TPM3 29 P49368 T-complex protein 1 subunit gamma CCT3 29 P236344 Plasma membrane calcium-transporting ATPase 4 ATP2B4 28 P11142 Heat shock cognate 71 kDa protein HSPA8 28 P60709 Actin, cytoplasmic 1; Actin, cytoplasmic 1,

N-terminally processed ACTB 28 Q5T4B7 E3 ubiquitin-protein ligase UBR4 27 P78371 T-complex protein 1 subunit beta CCT2 26 P28289 Tropomodulin-1 TMOD1 26 P50395 Rab GDP dissociation inhibitor beta GDI2 25 P27105 Erythrocyte band 7 integral membrane protein STOM 25 P68871 Hemoglobin subunit beta; LW-hemorphin-7; Spinorphin HBB 25 P02730 Band 3 anion transport protein SLC4A1 25 Q00013 55 kDa erythrocyte membrane protein MPP1 24 J3KPS3 Fructose-bisphosphate aldolase; Fructose-bisphosphate aldolase A ALDOA 24 P00352 Retinal dehydrogenase 1 ALDH1A1 24 Q86VP6 Cullin-associated NEDD8-dissociated protein 1 CAND1 24 P49327 Fatty acid synthase FASN 24 Q13228 Selenium-binding protein 1 SELENBP1 22 P00915 Carbonic anhydrase 1 CA1 22 P50991 T-complex protein 1 subunit delta CCT4 22 P50990 T-complex protein 1 subunit theta CCT8 22 P32119 Peroxiredoxin-2 PRDX2 21 P50995 Annexin A11 ANXA11 21 P69905 Hemoglobin subunit alpha HBA1 21 P00558 Phosphoglycerate kinase 1 PGK1 20 H7BXX9 ATP-binding cassette sub-family B member 6, mitochondrial ABCB6 20 P08758 Annexin A5; Annexin ANXA5 20 P09525 Annexin A4; Annexin ANXA4 20 P07900 Heat shock protein HSP 90-alpha HSP90AA1 20 P48643 T-complex protein 1 subunit epsilon CCT5 20 P07384 Calpain-1 catalytic subunit CAPN1 20 P22314 Ubiquitin-like modifier-activating enzyme 1 UBA1 19 P04406 Glyceraldehyde-3-phosphate dehydrogenase GAPDH 19 Q08495 Dematin DMTN 19 Q99832 T-complex protein 1 subunit eta CCT7 19 P29144 Tripeptidyl-peptidase 2 TPP2 19 P30041 Peroxiredoxin-6 PRDX6 18 E9PM69 26S protease regulatory subunit 6A PSMC3 18 P40227 T-complex protein 1 subunit zeta CCT6A 18 P50570 Dynamin-2 DNM2 18 E7EQB2 Lactotransferrin; Lactoferricin-H; Kaliocin-1; Lactoferroxin-A; Lactoferroxin-B; Lactoferroxin-C LTF 18 P31948 Stress-induced-phosphoprotein 1 STIP1 18 E7ESC6 Exportin-7 XPO7 18 E7EV99 Alpha-adducin ADD1 17 P62258 14-3-3 protein epsilon YWHAE 17 P30613 Pyruvate kinase PKLR PKLR 17 A0A0G2JIW1 Heat shock 70 kDa protein 1B; Heat shock 70 kDa protein 1A HSPA1B; 17 HSPA1A Q16531 DNA damage-binding protein 1 DDB1 17 P11021 78 kDa glucose-regulated protein HSPA5 17 F5H2F4 C-1-tetrahydrofolate synthase, cytoplasmic; Methylenetetrahydrofolate dehydrogenase MTHFD1 17 P07195 L-lactate dehydrogenase B chain; L-lactate dehydrogenase LDHB 16 P45974 Ubiquitin carboxyl-terminal hydrolase 5 USP5 16 O43242 26S proteasome non-ATPase regulatory subunit 3 PSMD3 16 C9J0K6 Sorcin SRI 16 P08133 Annexin A6; Annexin ANXA6 16 Q13200 26S proteasome non-ATPase regulatory subunit 2 PSMD2 16 P23276 Kell blood group glycoprotein KEL 16 P34932 Heat shock 70 kDa protein 4 HSPA4 16 A0A0A0MSI0 Peroxiredoxin-1 PRDX1 16 Q9Y230 RuvB-like 2 RUVBL2 16 Q5XPI4 E3 ubiquitin-protein ligase RNF123 RNF123 16 P68871 Hemoglobin subunit beta; LW-hemorphin-7; Spinorphin HBB 15 P60174 Triosephosphate isomerase TPI1 15 P00491 Purine nucleoside phosphorylase PNP 15 C9JIF9 Acylamino-acid-releasing enzyme APEH 15 H7BYY1 Tropomyosin alpha-1 chain TPM1 15 P35998 26S protease regulatory subunit 7 PSMC2 15 P17987 T-complex protein 1 subunit alpha TCP1 15 P09543 2,3-cyclic-nucleotide 3-phosphodiesterase CNP 15 Q99460 26S proteasome non-ATPase regulatory subunit 1 PSMD1 15 Q9Y4E8 Ubiquitin carboxyl-terminal hydrolase 15 USP15 15 Q9C0C9 E2/E3 hybrid ubiquitin-protein ligase UBE2O UBE2O 15 P26038 Moesin MSN 15 P04083 Annexin A1; Annexin ANXA1 14 P30043 Flavin reductase (NADPH) BLVRB 14 P11166 Solute carrier family 2, facilitated glucose transporter member 1 SLC2A1 14 P00918 Carbonic anhydrase 2 CA2 14 P06733 Alpha-enolase ENO1 14 Q5TDH0 Protein DDI1 homolog 2 DDI2 14 B0QZ18 Copine-1 CPNE1 14 O75326 Semaphorin-7A SEMA7A 14 P05164 Myeloperoxidase MPO 14 Q9Y265 RuvB-like 1 RUVBL1 14 P29401 Transketolase TKT 14 I3L0N3 Vesicle-fusing ATPase NSF 14 Q4VB86 Protein 4.1 EPB41 14 P11277 Spectrin beta chain, erythrocytic SPTB 14 P13716 Delta-aminolevulinic acid dehydratase ALAD 13 P07738 Bisphosphoglycerate mutase BPGM 13 P48506 Glutamate-cysteine ligase catalytic subunit GCLC 13 Q99816 Tumor susceptibility gene 101 protein TSG101 13 O14818 Proteasome subunit alpha type-7 PSMA7 13 P23526 Adenosylhomocysteinase AHCY 13 P61225 Ras-related protein Rap-2b RAP2B 13 O00231 26S proteasome non-ATPase regulatory subunit 11 PSMD11 13 P11413 Glucose-6-phosphate 1-

dehydrogenase G6PD13 P00338 L-lactate dehydrogenase A chain LDHA 12 Q99808 Equilibrative nucleoside transporter 1 SLC29A1 12 A6NJA2 Ubiquitin carboxyl-terminal hydrolase; Ubiquitin carboxyl-terminal hydrolase 14 USP14 12 Q06323 Proteasome activator complex subunit 1 PSME1 12 P28074 Proteasome subunit beta type-5 PSMB5 12 B3KQV6 Serine/threonine-protein phosphatase 2A 65 kDa regulatory subunit A alpha isoform PPP2R1A 12 Q14974 Importin subunit beta-1 KPNB1 12 P25786 Proteasome subunit alpha type-1; Proteasome subunit alpha type PSMA1 12 Q86X55 Histone-arginine methyltransferase CARM1 CARM1 12 A6NG10 WW domain-binding protein 2 WBP2 12 P63092 Guanine nucleotide-binding protein G(s) subunit alpha GNAS 12 P31939 Bifunctional purine biosynthesis protein PURH; Phosphoribosylaminoimidazolecarboxamide ATIC 12 formyltransferase; IMP cyclohydrolase P52209 6-phosphogluconate dehydrogenase, decarboxylating PGD 12 A0A087X0C8 Calpain-5 CAPN5 12 F8W9S7 GTPase-activating protein and VPS9 domain-containing protein 1 GAPVD1 12 P60842 Eukaryotic initiation factor 4A-I EIF4A1 12 P69905 Hemoglobin subunit alpha HBA1 11 Q6XQN6 Nicotinate phosphoribosyltransferase NAPRT 11 P48637 Glutathione synthetase GSS 11 H7BZ94 Protein disulfide-isomerase P4HB 11 P21980 Protein-glutamine gamma-glutamyltransferase 2 TGM2 11 P50895 Basal cell adhesion molecule BCAM 11 A0A087X2I1 26S protease regulatory subunit 10B PSMC6 11 G3V1D3 Dipeptidyl peptidase 3 DPP3 11 P05023 Sodium/potassium-transporting ATPase subunit alpha-1; Sodium/potassium-transporting ATP1A1; ATP1A3 11 ATPase subunit alpha-3 P40925 Malate dehydrogenase, cytoplasmic; Malate dehydrogenase MDH1 11 Q9UKV8 Protein argonaute-2 AGO2 11 P30566 Adenylosuccinate lyase ADSL 11 P20618 Proteasome subunit beta type-1 PSMB1 11 P17858 ATP-dependent 6-phosphofructokinase, liver type PFKL 11 A0A087X253 AP-2 complex subunit beta AP2B1 11 O95782 AP-2 complex subunit alpha-1 AP2A1 11 O00232 26S proteasome non-ATPase regulatory subunit 12 PSMD12 11 Q9BSL1 Ubiquitin-associated domain-containing protein 1 UBAC1 11 A0A087WUL0 Bifunctional ATP-dependent dihydroxyacetone kinase/FAD-AMP lyase (cyclizing); ATP- TKFC; DAK 11 dependent dihydroxyacetone kinase; FAD-AMP lyase (cyclizing) P69891 Hemoglobin subunit gamma-1 HBG1 11 A0A087WZE4 Spectrin alpha chain, erythrocytic 1 SPTA1 11 P48426 Phosphatidylinositol 5-phosphate 4-kinase type-2 alpha PIP4K2A 10 P10644 cAMP-dependent protein kinase type I-alpha regulatory subunit PRKAR1A 10 Q9BWD1 Acetyl-CoA acetyltransferase, cytosolic ACAT2 10 P62191 26S protease regulatory subunit 4 PSMC1 10 E9PBS1 Multifunctional protein ADE2; Phosphoribosylaminoimidazole-succinocarboxamide PAICS 10 synthase; Phosphoribosylaminoimidazole carboxylase M0R0Y2 Alpha-soluble NSF attachment protein NAPA 10 P78417 Glutathione S-transferase omega-1 GSTO1 10 P25789 Proteasome subunit alpha type-4; Proteasome subunit alpha type; Proteasome subunit beta PSMA4 10 type Q9H0U4 Ras-related protein Rab-1B; Putative Ras-related protein Rab-1C RAB1B; RAB1C 10 Q16401 26S proteasome non-ATPase regulatory subunit 5 PSMD5 10 P30101 Protein disulfide-isomerase A3 PDIA3 10 D6RAX7 COP9 signalosome complex subunit 4 COPS4 10 O75340 Programmed cell death protein 6 PDCD6 10 Q96P70 Importin-9 IPO9 10 P38606 V-type proton ATPase catalytic subunit A ATP6V1A 10 A0A0G2JH68 Protein diaphanous homolog 1 DIAPH1 10 Q5T9B7 Adenylate kinase isoenzyme 1 AK1 10 O14980 Exportin-1 XPO1 10 P43686 26S protease regulatory subunit 6B PSMC4 10 H0YH81 ATP synthase subunit beta; ATP synthase subunit beta, mitochondrial ATP5B 10 Q04656 Copper-transporting ATPase 1 ATP7A 10 P16452 Erythrocyte membrane protein band 4.2 EPB42 10 Q9BY43 Charged multivesicular body protein 4a CHMP4A 9 P51148 Ras-related protein Rab-5C RAB5C 9 Q9NRV9 Heme-binding protein 1 HEBP1 9 Q9UNZ2 NSFL1 cofactor p47 NSFL1C 9 Q16851 UTP--glucose-1-phosphate uridylyltransferase UGP2 9 P28066 Proteasome subunit alpha type-5 PSMA5 9 A0A0C4DGQ5 Calpain small subunit 1 CAPNS1 9 A0A087X1Z3 Proteasome activator complex subunit 2 PSME2 9 Q01518 Adenylyl cyclase-associated protein 1 CAP1 9 B1AKQ8 Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-1 GNB1 9 O75131 Copine-3 CPNE3 9 P54725 UV excision repair protein RAD23 homolog A RAD23A 9 P11215 Integrin

alpha-M ITGAM 9 Q93008 Probable ubiquitin carboxyl-terminal hydrolase FAF-X USP9X 9
Q96G03 Phosphoglucosyltransferase-2 PGM2 9 P49721 Proteasome subunit beta type-2 PSMB2 9
Q15008 26S proteasome non-ATPase regulatory subunit 6 PSMD6 9 Q9UNQ0 ATP-binding
cassette sub-family G member 2 ABCG2 9 P22303 Acetylcholinesterase; Carboxylic ester
hydrolase ACHE 9 G3V5Z7 Proteasome subunit alpha type; Proteasome subunit alpha type-6
PSMA6 9 O15439 Multidrug resistance-associated protein 4 ABCC4 9 P37837 Transaldolase
TALDO1 9 O14744 Protein arginine N-methyltransferase 5 PRMT5 9 P02042 Hemoglobin subunit
delta HBD 9 P25788 Proteasome subunit alpha type-3 PSMA3 8 J3QS39 Ubiquitin-60S ribosomal
protein L40; Ubiquitin; 60S ribosomal protein L40; Ubiquitin-40S UBB; RPS27A; 8 ribosomal
protein S27a; Ubiquitin; 40S ribosomal protein S27a; Polyubiquitin- UBC; UBA52; B; Ubiquitin;
Polyubiquitin-C; Ubiquitin UBBP4 O94919 Endonuclease domain-containing 1 protein ENDOD1
8 P31946 14-3-3 protein beta/alpha YWHAB 8 P60891 Ribose-phosphate pyrophosphokinase 1;
Ribose-phosphate pyrophosphokinase 2; Ribose- PRPS1; PRPS2; 8 phosphate pyrophosphokinase
3 PRPS1L1 P62195 26S protease regulatory subunit 8 PSMC5 8 A0A024RA52 Proteasome
subunit alpha type; Proteasome subunit alpha type-2 PSMA2 8 P04899 Guanine nucleotide-binding
protein G(i) subunit alpha-2 GNAI2 8 Q13561 Dynactin subunit 2 DCTN2 8 P53004 Biliverdin
reductase A BLVRA 8 P00387 NADH-cytochrome b5 reductase 3 CYB5R3 8 P16152 Carbonyl
reductase [NADPH] 1 CBR1 8 P06744 Glucose-6-phosphate isomerase GPI 8 Q99733
Nucleosome assembly protein 1-like 4 NAP1L4 8 P04792 Heat shock protein beta-1 HSPB1 8
P17612 cAMP-dependent protein kinase catalytic subunit alpha; cAMP-dependent protein kinase
PRKACA; KIN27; 8 catalytic subunit beta PRKACB P20340 Ras-related protein Rab-6A RAB6A
8 P13796 Plastin-2 LCP1 8 P52907 F-actin-capping protein subunit alpha-1 CAPZA1 8 Q14697
Neutral alpha-glucosidase AB GANAB 8 P08514 Integrin alpha-IIb; Integrin alpha-IIb heavy
chain; Integrin alpha-IIb light chain, form 1; Integrin ITGA2B 8 alpha-IIb light chain, form 2
P26641 Elongation factor 1-gamma EEF1G 8 Q9UQ80 Proliferation-associated protein 2G4
PA2G4 8 Q9Y4D1 Disheveled-associated activator of morphogenesis 1 DAAM1 8 P11166 Solute
carrier family 2, facilitated glucose transporter member 1 SLC2A1 8 P63261 Actin, cytoplasmic 2;
Actin, cytoplasmic 2, N-terminally processed; Actin, cytoplasmic 1; Actin, ACTG1; ACTB; 8
cytoplasmic 1, N-terminally processed; Actin, gamma-enteric smooth muscle; Actin, alpha
ACTG2; ACTA1; skeletal muscle; Actin, alpha cardiac muscle 1; Actin, aortic smooth muscle
ACTC1; ACTA2 P02008 Hemoglobin subunit zeta HBZ 8 P20073 Annexin A7 ANXA7 8 P35613
Basigin BSG 7 P49720 Proteasome subunit beta type-3 PSMB3 7 P17931 Galectin-3; Galectin
LGALS3 7 P63104 14-3-3 protein zeta/delta YWHAZ 7 Q9H444 Charged multivesicular body
protein 4b CHMP4B 7 O43396 Thioredoxin-like protein 1 TXNL1 7 P50502 Hsc70-interacting
protein; Putative protein FAM10A4; Putative protein FAM10A5 ST13; ST13P4; 7 ST13P5
E7EQ12 Calpastatin CAST 7 P49189 4-trimethylaminobutyraldehyde dehydrogenase ALDH9A1 7
Q9Y3I1 F-box only protein 7 FBXO7 7 P07954 Fumarate hydratase, mitochondrial FH 7 F6S8N6
Protein-L-isoaspartate O-methyltransferase; Protein-L-isoaspartate(D-aspartate) O- PCMT1 7
methyltransferase P49247 Ribose-5-phosphate isomerase RPIA 7 P62834 Ras-related protein Rap-
1A RAP1A 7 A0A087WUQ6 Glutathione peroxidase; Glutathione peroxidase 1 GPX1 7 O60256
Phosphoribosyl pyrophosphate synthase-associated protein 2 PRPSAP2 7 Q00796 Sorbitol
dehydrogenase SORD 7 O00299 Chloride intracellular channel protein 1 CLIC1 7 Q99497 Protein
deglycase DJ-1 PARK7 7 P17174 Aspartate aminotransferase, cytoplasmic GOT1 7 H7BXD5
Grancalcin GCA 7 E9PGT1 Translin TSN 7 C9J7K9 Phospholipid scramblase 1 PLSCR1 7
Q13618 Cullin-3 CUL3 7 O75695 Protein XRP2 RP2 7 P09960 Leukotriene A-4 hydrolase LTA4H
7 E9PLK3 Puromycin-sensitive aminopeptidase NPEPPS 7 P00492 Hypoxanthine-guanine
phosphoribosyltransferase HPRT1 7 F5H4B6 Aldehyde dehydrogenase family 16 member A1
ALDH16A1 7 D6RA82 Annexin; Annexin A3 ANXA3 7 P61106 Ras-related protein Rab-14
RAB14 7 Q16775 Hydroxyacylglutathione hydrolase, mitochondrial HAGH 7 P07355 Annexin
A2; Annexin; Putative annexin A2-like protein ANXA2; ANXA2P2 7 A0A087WX08 Gamma-

adducin ADD3 7 P08238 Heat shock protein HSP 90-beta HSP90AB1 7 Q8IZY2 ATP-binding cassette sub-family A member 7 ABCA7 7 P14780 Matrix metalloproteinase-9 MMP9 7 P12955 Xaa-Pro dipeptidase PEPD 7 P68371 Tubulin beta-4B chain; Tubulin beta-4A chain; Tubulin beta chain TUBB4B; 7 TUBB4A; TUBB O15067 Phosphoribosylformylglycinamide synthase PFAS 7 P35241 Radixin RDX 7 O60488 Long-chain-fatty-acid--CoA ligase 4 ACSL4 7 A0A0C4DGX4 Cullin-1 CUL1 7 P50148 Guanine nucleotide-binding protein G(q) subunit alpha GNAQ 7 Q15907 Ras-related protein Rab-11B; Ras-related protein Rab-11A RAB11B; 7 RAB11A P30086 Phosphatidylethanolamine-binding protein 1; Hippocampal cholinergic neurostimulating PEBP1 6 peptide P55036 26S proteasome non-ATPase regulatory subunit 4 PSMD4 6 P28070 Proteasome subunit beta type-4 PSMB4 6 Q9UNS2 COP9 signalosome complex subunit 3 COPS3 6 P08754 Guanine nucleotide-binding protein G(k) subunit alpha GNAI3 6 P61006 Ras-related protein Rab-8A RAB8A 6 P61019 Ras-related protein Rab-2A RAB2A 6 H0Y8C6 Importin-5 IPO5 6 Q00577 Transcriptional activator protein Pur-alpha PURA 6 P52565 Rho GDP-dissociation inhibitor 1 ARHGDIA 6 Q9Y5Z4 Heme-binding protein 2 HEBP2 6 J3KNF4 Copper chaperone for superoxide dismutase; Superoxide dismutase [Cu—Zn] CCS 6 A0A087WXS7 ATPase ASNA1 ASNA1 6 P05089 Arginase-1 ARG1 6 O95336 6-phosphogluconolactonase PGLS 6 Q92508 Piezo-type mechanosensitive ion channel component 1 PIEZO1 6 P84077 ADP-ribosylation factor 1; ADP-ribosylation factor 3 ARF1; ARF3 6 P21281 V-type proton ATPase subunit B, brain isoform; V-type proton ATPase subunit B, kidney ATP6V1B2; 6 isoform ATP6V1B1 P04259 Keratin, type II cytoskeletal 6B KRT6B 6 Q92905 COP9 signalosome complex subunit 5 COPS5 6 P61163 Alpha-actinin ACTR1A 6 O95373 Importin-7 IPO7 6 C9JD73 Protein phosphatase 1 regulatory subunit 7 PPP1R7 6 Q99536 Synaptic vesicle membrane protein VAT-1 homolog VAT1 6 Q86UX7 Fermitin family homolog 3 FERMT3 6 C9JFE4 COP9 signalosome complex subunit 1 GPS1 6 P36959 GMP reductase 1 GMPR 6 B5MDF5 GTP-binding nuclear protein Ran RAN 6 F5GY90 Porphobilinogen deaminase HMBS 6 E7EX90 Dynactin subunit 1 DCTN1 6 H0Y512 Adipocyte plasma membrane-associated protein APMAP 6 Q32Q12 Nucleoside diphosphate kinase; Nucleoside diphosphate kinase B; Putative nucleoside NME1-NME2; 6 diphosphate kinase NME2; NME1; NME2P1 P13807 Glycogen [starch] synthase, muscle GYS1 6 K7ES02 Bleomycin hydrolase BLMH 6 P17213 Bactericidal permeability-increasing protein BPI 6 Q9UNM6 26S proteasome non-ATPase regulatory subunit 13 PSMD13 6 J3KQ32 Obg-like ATPase 1 OLA1 6 Q9Y490 Talin-1 TLN1 6 H0YD13 CD44 antigen CD44 6 P18669 Phosphoglycerate mutase 1; Phosphoglycerate mutase 2; Probable phosphoglycerate PGAM1; PGAM2; 6 mutase 4 PGAM4 P23528 Cofilin-1 CFL1 6 Q5SR44 Complement receptor type 1 CR1 6 Q99436 Proteasome subunit beta type-7 PSMB7 6 P47756 F-actin-capping protein subunit beta CAPZB 6 P30740 Leukocyte elastase inhibitor SERPINB1 6 A0A024R571 EH domain-containing protein 1 EHD1 6 P30043 Flavin reductase (NADPH) BLVRB 6 P32119 Peroxiredoxin-2 PRDX2 6 C9J0K6 Sorcin SRI 6 P27105 Erythrocyte band 7 integral membrane protein STOM 6 P84077 ADP-ribosylation factor 1; ADP-ribosylation factor 3; ADP-ribosylation factor 5; ADP- ARF1; ARF3; 6 ribosylation factor 4 ARF5; ARF4 P61981 14-3-3 protein gamma; 14-3-3 protein gamma, N-terminally processed YWHAG 5 P09211 Glutathione S-transferase P GSTP1 5 P13489 Ribonuclease inhibitor RNH1 5 Q96PU5 E3 ubiquitin-protein ligase NEDD4-like NEDD4L 5 Q5SRN7 HLA class I histocompatibility antigen, A; HLA class I histocompatibility antigen, B; HLA HLA-A; 5 class I histocompatibility antigen, Cw HLA-C; HLA-B P02042 Hemoglobin subunit delta HBD 5 A0A087WU29 Glycophorin-A GYP A 5 P00390 Glutathione reductase, mitochondrial GSR 5 Q9UBV8 Peflin PEF1 5 Q8WVM8 Sec1 family domain-containing protein 1 SCFD1 5 F6TLX2 Glyoxalase domain-containing protein 4 GLOD4 5 H3BQF1 Adenine phosphoribosyltransferase APRT 5 Q07960 Rho GTPase-activating protein 1 ARHGAP1 5 F6XSS0 Blood group Rh(CE) polypeptide; Blood group Rh(D) polypeptide RHCE; RHD 5 A0A087WY55 Vacuolar protein sorting-associated protein VTA1 homolog VTA1 5 G5E9W8 Glycogenin-1 GYG1 5 P07451 Carbonic anhydrase 3 CA3 5 Q08722 Leukocyte surface antigen CD47 CD47 5 X6RA14 S-

formylglutathione hydrolase ESD 5 Q8IU8 Cytokine receptor-like factor 3 CRLF3 5 Q5VW32
 BRO1 domain-containing protein BROX BROX 5 P61026 Ras-related protein Rab-10 RAB10 5
 A0A087WWY3 Filamin-A FLNA 5 H0YGX7 Rho GDP-dissociation inhibitor 2 ARHGDIB 5
 Q92783 Signal transducing adapter molecule 1 STAM 5 Q7Z6Z7 E3 ubiquitin-protein ligase
 HUWE1 HUWE1 5 H0YHC3 Nucleosome assembly protein 1-like 1 NAP1L1 5 Q5QPM7
 Proteasome inhibitor PI31 subunit PSMF1 5 P09104 Gamma-enolase; Enolase ENO2 5 Q5T2B5
 Cullin-2 CUL2 5 Q8WW22 DnaJ homolog subfamily A member 4 DNAJA4 5 P61201 COP9
 signalosome complex subunit 2 COPS2 5 X6R433 Protein-tyrosine-phosphatase; Receptor-type
 tyrosine-protein phosphatase C PTPRC 5 P63000 Ras-related C3 botulinum toxin substrate 1
 RAC1 5 F5GXM3 IST1 homolog IST1 5 H3BLU7 Aflatoxin B1 aldehyde reductase member 2
 AKR7A2 5 P25325 3-mercaptopyruvatesulfurtransferase; Sulfurtransferase MPST 5 A0A087X0K1
 Calcium-binding protein 39 CAB39 5 P23381 Tryptophan--tRNA ligase, cytoplasmic; T1-TrpRS;
 T2-TrpRS WARS 5 P01116 GTPase KRas; GTPase KRas, N-terminally processed KRAS 5
 P30040 Endoplasmic reticulum resident protein 29 ERP29 5 P05198 Eukaryotic translation
 initiation factor 2 subunit 1 EIF2S1 5 Q6UX06 Olfactomedin-4 OLFM4 5 Q96KP4 Cytosolic non-
 specific dipeptidase CNDP2 5 Q04760 Lactoylglutathione lyase GLO1 5 J3QS39 Ubiquitin-60S
 ribosomal protein L40; Ubiquitin; 60S ribosomal protein L40; Ubiquitin-40S UBB; RPS27A; 5
 ribosomal protein S27a; Ubiquitin; 40S ribosomal protein S27a; Polyubiquitin- UBC; UBA52; B;
 Ubiquitin; Polyubiquitin-C; Ubiquitin UBBP4 P06702 Protein S100-A9 S100A9 5 P15531
 Nucleoside diphosphate kinase A; Nucleoside diphosphate kinase; Nucleoside diphosphate NME1;
 NME2; 5 kinase B NME1-NME2 E7EV99 Alpha-adducin ADD1 5 P51149 Ras-related protein
 Rab-7a RAB7A 4 K7N7A8 Aquaporin-1 AQP1 4 P61020 Ras-related protein Rab-5B RAB5B 4
 X6R4N5 Erythroid membrane-associated protein ERMAP 4 E5RJR5 S-phase kinase-associated
 protein 1 SKP1 4 Q9Y315 Deoxyribose-phosphate aldolase DERA 4 X6R8F3 Neutrophil
 gelatinase-associated lipocalin LCN2 4 O75396 Vesicle-trafficking protein SEC22b SEC22B 4
 Q15102 Platelet-activating factor acetylhydrolase IB subunit gamma PAFAH1B3 4 P51665 26S
 proteasome non-ATPase regulatory subunit 7 PSMD7 4 Q96FZ7 Charged multivesicular body
 protein 6 CHMP6 4 Q9NRQ2 Phospholipid scramblase 4 PLSCR4 4 F8VWS0 60S acidic
 ribosomal protein P0; 60S acidic ribosomal protein P0-like RPLP0; 4 RPLP0P6 O14964
 Hepatocyte growth factor-regulated tyrosine kinase substrate HGS 4 A0A0A0MTJ9 Neutral
 cholesterol ester hydrolase 1 NCEH1 4 A0A087WY82 Junctional adhesion molecule A F11R 4
 J3QSB7 Purine nucleoside phosphorylase; S-methyl-5-thioadenosine phosphorylase MTAP 4
 Q5WQ6 Ubiquitin thioesterase OTU1 YOD1 4 P36543 V-type proton ATPase subunit E 1
 ATP6V1E1 4 Q9BS40 Latexin LXN 4 P47755 F-actin-capping protein subunit alpha-2 CAPZA2 4
 Q9GZT8 NIF3-like protein 1 NIF3L1 4 H6UYS7 Alpha-synuclein SNCA 4 P60953 Cell division
 control protein 42 homolog CDC42 4 Q14773 Intercellular adhesion molecule 4 ICAM4 4
 H0Y6T7 Nicastrin NCSTN 4 P69891 Hemoglobin subunit gamma-1 HBG1 4 P14625
 Endoplasmin HSP90B1 4 Q96GD0 Pyridoxal phosphate phosphatase PDXP 4 P08311 Cathepsin G
 CTSG 4 Q9H9Q2 COP9 signalosome complex subunit 7b COPS7B 4 P09417 Dihydropteridine
 reductase QDPR 4 F8WE6 Peptidyl-prolyl cis-trans isomerase; Peptidyl-prolyl cis-trans isomerase
 A PPIA 4 O15173 Membrane-associated progesterone receptor component 2 PGRMC2 4 P28072
 Proteasome subunit beta type-6; Proteasome subunit beta type PSMB6 4 H3BSW0 Leucine-rich
 repeat-containing protein 57 LRRC57 4 Q9UBW8 COP9 signalosome complex subunit 7a
 COPS7A 4 O00560 Syntenin-1 SDCBP 4 F5H157 Ras-related protein Rab-35 RAB35 4 Q9H479
 Fructosamine-3-kinase FN3K 4 Q04917 14-3-3 protein eta YWHAH 4 C9JJ47 AP-2 complex
 subunit mu AP2M1 4 C9JIG9 Serine/threonine-protein kinase OSR1 OXSR1 4 Q13336 Urea
 transporter 1 SLC14A1 4 P46926 Glucosamine-6-phosphate isomerase 1; Glucosamine-6-
 phosphate GNPDA1; 4 isomerase; Glucosamine-6-phosphate isomerase 2 GNPDA2 O43633
 Charged multivesicular body protein 2a CHMP2A 4 F8VVB9 Tubulin alpha-1A chain; Tubulin
 alpha-1C chain; Tubulin alpha-1B chain; Tubulin alpha-3C/D TUBA1B; TUBA1C; 4 chain;

Tubulin alpha-3E chain TUBA1A; TUBA1A; TUBA3E A6PVN5 Serine/threonine-protein phosphatase 2A activator PPP2R4 4 B8ZZB8 CB1 cannabinoid receptor-interacting protein 1 CNRIP1 4 Q9P2R3 Rabankyrin-5 ANKFY1 4 Q86YS7 C2 domain-containing protein 5 C2CD5 4 R4GMR5 26S proteasome non-ATPase regulatory subunit 8 PSMD8 4 E7EM64 COP9 signalosome complex subunit 6 COPS6 4 J3KNI6 Integrin beta; Integrin beta-2 ITGB2 4 H0Y5R6 Uroporphyrinogen decarboxylase UROD 4 M0R165 Epidermal growth factor receptor substrate 15-like 1 EPS15L1 4 O00487 26S proteasome non-ATPase regulatory subunit 14 PSMD14 4 Q13630 GDP-L-fucose synthase TSTA3 4 P55060 Exportin-2 CSE1L 4 P20020 Plasma membrane calcium-transporting ATPase 1; Calcium-transporting ATPase ATP2B1 4 Q9NYU2 UDP-glucose: glycoprotein glucosyltransferase 1 UGGT1 4 H3BND8 Ubiquitin carboxyl-terminal hydrolase; Ubiquitin carboxyl-terminal hydrolase 7 USP7 4 Q9GZP4 PITH domain-containing protein 1 PITHD1 4 D6RD66 WD repeat-containing protein 1 WDR1 4 P48729 Casein kinase I isoform alpha CSNK1A1 4 P25685 DnaJ homolog subfamily B member 1 DNAJB1 4 P14550 Alcohol dehydrogenase [NADP(+)] AKR1A1 4 Q6PCE3 Glucose 1,6-bisphosphate synthase PGM2L1 4 Q9UPN7 Serine/threonine-protein phosphatase 6 regulatory subunit 1 PPP6R1 4 P62805 Histone H4 HIST1H4A 4 P62937 Peptidyl-prolyl cis-trans isomerase A PPIA 4 P30046 D-dopachrome decarboxylase; D-dopachrome decarboxylase-like protein DDT; DDTL 4 Q08495 Dematin DMTN 4 K7EIJ0 WW domain-binding protein 2 WBP2 4 E5RHP7 Carbonic anhydrase 1 CA1 4 P09105 Hemoglobin subunit theta-1 HBQ1 4 E7ESC6 Exportin-7 XPO7 4 P61225 Ras-related protein Rap-2b; Ras-related protein Rap-2c; Ras-related protein Rap-2a RAP2B; RAP2A; 4 RAP2C O15400 Syntaxin-7 STX7 3 Q5VZR0 Golgi-associated plant pathogenesis-related protein 1 GLIPR2 3 P40199 Carcinoembryonic antigen-related cell adhesion molecule 6 CEACAM6 3 P27348 14-3-3 protein theta YWHAG 3 P62820 Ras-related protein Rab-1A RAB1A 3 C9JEN3 Protein lifeguard 3 TMBIM1 3 Q9UDX3 SEC14-like protein 4 SEC14L4 3 Q9Y570 Protein phosphatase methylesterase 1 PPME1 3 Q96GG9 DCN1-like protein 1; DCN1-like protein DCUN1D1 3 B0YJC4 Vimentin VIM 3 Q86VN1 Vacuolar protein-sorting-associated protein 36 VPS36 3 F8WFB9 Endophilin-B2 SH3GLB2 3 A0A087WVQ9 Elongation factor 1-alpha 1; Putative elongation factor 1-alpha-like 3 EEF1A1; 3 EEF1A1P5 Q53TN4 Cytochrome b reductase 1 CYBRD1 3 P10809 60 kDa heat shock protein, mitochondrial HSPD1 3 F5H7X1 26S proteasome non-ATPase regulatory subunit 9 PSMD9 3 E9PNW4 CD59 glycoprotein CD59 3 E9PS74 Solute carrier family 43 member 3 SLC43A3 3 P53985 Monocarboxylate transporter 1 SLC16A1 3 Q8WWI5 Choline transporter-like protein 1 SLC44A1 3 E9PIR7 Thioredoxin reductase 1, cytoplasmic TXNRD1 3 Q5TD07 Ribosyldihydronicotinamide dehydrogenase [quinone] NQO2 3 Q9UL25 Ras-related protein Rab-21 RAB21 3 P27824 Calnexin CANX 3 U3KPS2 Myeloblastin PRTN3 3 Q9UKU0 Long-chain-fatty-acid--CoA ligase 6 ACSL6 3 P24666 Low molecular weight phosphotyrosine protein phosphatase ACP1 3 H7C2G2 NAD(P)(+)--arginine ADP-ribosyltransferase; Ecto-ADP-ribosyltransferase 4 ART4 3 I3L1K6 Myosin light chain 4 MYL4 3 Q9UK41 Vacuolar protein sorting-associated protein 28 homolog VPS28 3 Q9NUQ9 Protein FAM49B FAM49B 3 Q9UBQ7 Glyoxylate reductase/hydroxypyruvate reductase GRHPR 3 B8ZZG1 MAGUK p55 subfamily member 6 MPP6 3 Q10567 AP-1 complex subunit beta-1 AP1B1 3 O75387 Large neutral amino acids transporter small subunit 3 SLC43A1 3 Q9BTU6 Phosphatidylinositol 4-kinase type 2-alpha PI4K2A 3 J3KS22 L-xylulose reductase DCXR 3 O15498 Synaptobrevin homolog YKT6 YKT6 3 Q08211 ATP-dependent RNA helicase A DHX9 3 P50416 Carnitine O-palmitoyltransferase 1, liver isoform CPT1A 3 P08237 ATP-dependent 6-phosphofructokinase, muscle type PFKM 3 P14735 Insulin-degrading enzyme IDE 3 Q9H0R3 Transmembrane protein 222 TMEM222 3 Q14166 Tubulin--tyrosine ligase-like protein 12 TTLL12 3 Q14558 Phosphoribosyl pyrophosphate synthase-associated protein 1 PRPSAP1 3 A0A087WTB8 Ubiquitin carboxyl-terminal hydrolase; Ubiquitin carboxyl-terminal hydrolase isoform L3 UCHL3 3 Q9NPQ8 Synembryn-A RIC8A 3 F6WQW2 Ran-specific GTPase-activating protein RANBP1 3 Q15691 Microtubule-associated protein RP/EB family member 1

MAPRE1 3 A0A00MR50 Cullin-4A CUL4A 3 Q96IU4 Alpha/beta hydrolase domain-containing protein 14B ABHD14B 3 Q9P0L0 Vesicle-associated membrane protein-associated protein A VAPA 3 E9PRY8 Elongation factor 1-delta EEF1D 3 Q16543 Hsp90 co-chaperone Cdc37 CDC37 3 P06702 Protein S100-A9 S100A9 3 P13639 Elongation factor 2 EEF2 3 E9PJC7 Tetraspanin; CD82 antigen CD82 3 Q01432 AMP deaminase 3 AMPD3 3 B1AUU8 Epidermal growth factor receptor substrate 15 EPS15 3 P54709 Sodium/potassium-transporting ATPase subunit beta-3 ATP1B3 3 P54727 UV excision repair protein RAD23 homolog B RAD23B 3 P36507 Dual specificity mitogen-activated protein kinase kinase 2 MAP2K2 3 Q9Y2V2 Calcium-regulated heat stable protein 1 CARHSP1 3 P31146 Coronin-1A; Coronin CORO1A 3 Q8IY17 Neuropathy target esterase PNPLA6 3 E7EQR4 Ezrin EZR 3 F8WDS9 LanC-like protein 1 LANCL1 3 X6RJP6 Transgelin-2 TAGLN2 3 A0A0J9YXM6 WD repeat-containing protein 81 WDR81 3 P62140 Serine/threonine-protein phosphatase PP1-beta catalytic subunit; Serine/threonine-protein PPP1CB 3 phosphatase H0YLJ3 Mortality factor 4-like protein 1 MORF4L1 3 E7EPV7 Alpha-synuclein SNCA 3 K7EK07 Histone H3 H3F3B 3 P05109 Protein S100-A8; Protein S100-A8, N-terminally processed S100A8 3 I3L0A0 Ubiquitin-conjugating enzyme E2 variant 1; Ubiquitin-conjugating enzyme E2 variant 2 TMEM189- 3 UBE2V1; UBE2V1; UBE2V2 Q6B0K9 Hemoglobin subunit mu HBM 3 O75368 SH3 domain-binding glutamic acid-rich-like protein SH3BGRL 3 A0A0J9YXB3 Ras-related protein Rap-1b; Ras-related protein Rap-1A; Ras-related protein Rap-1b-like RAP1B; RAP1A 3 protein P60953 Cell division control protein 42 homolog CDC42 3 A0A087WT11 Ras-related protein Rab-1B; Ras-related protein Rab-1A RAB1B; RAB1A 3 B5MDF5 GTP-binding nuclear protein Ran RAN 3 P61088 Ubiquitin-conjugating enzyme E2 N UBE2N 3 Q00013 55 kDa erythrocyte membrane protein MPP1 3 B1AKQ8 Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-1; Guanine nucleotide- GNB1; GNB2; 3 binding protein G(I)/G(S)/G(T) subunit beta-3; Guanine nucleotide-binding protein GNB3 G(I)/G(S)/G(T) subunit beta-2 P04406 Glyceraldehyde-3-phosphate dehydrogenase GAPDH 3 Q9UJC5 SH3 domain-binding glutamic acid-rich-like protein 2 SH3BGRL2 3 P04921 Glycophorin-C GYPC 2 Q9NP59 Solute carrier family 40 member 1 SLC40A1 2 K7EKH5 Fructose-bisphosphate aldolase; Fructose-bisphosphate aldolase C ALDOC 2 P11233 Ras-related protein Ral-A RALA 2 F8WBR5 Calmodulin CALM2; 2 CALM3; CALM1 P00441 Superoxide dismutase [Cu—Zn] SOD1 2 H0YDI1 Lymphocyte function-associated antigen 3 CD58 2 H0YNE9 Ras-related protein Rab-8B RAB8B 2 J3KN67 TPM3 2 P20160 Azurocidin AZU1 2 A0A087WZZ4 Ammonium transporter Rh type A RHAG 2 F5GXS0 Complement C4-A; Complement C4-B C4B; C4A 2 Q8ND76 Cyclin-Y CCNY 2 Q14739 Lamin-B receptor LBR 2 Q15181 Inorganic pyrophosphatase PPA1 2 J3QS92 Galectin-9 LGALS9 2 I3L471 Phosphatidylinositol transfer protein alpha isoform PITPNA 2 P01111 GTPase NRas NRAS; KRAS 2 F5H4Q5 Vacuolar protein sorting-associated protein 37C VPS37C 2 A0A0A0MSW4 Phosphatidylinositol transfer protein beta isoform PITPNB 2 P67775 Serine/threonine-protein phosphatase 2A catalytic subunit alpha isoform; Serine/threonine-PPP2CA; 2 protein phosphatase PPP2CB P41091 Eukaryotic translation initiation factor 2 subunit 3; Putative eukaryotic translation initiation EIF2S3; 2 factor 2 subunit 3-like protein EIF2S3L A0A087WWS7 Syntaxin-binding protein 2 STXBP2 2 F8VQX6 Methyltransferase-like protein 7A METTL7A 2 B3KT28 FAS-associated factor 1 FAF1 2 K7EIJ8 Katanin p60 ATPase-containing subunit A-like 2 KATNAL2 2 P20339 Ras-related protein Rab-5A RAB5A 2 O95456 Proteasome assembly chaperone 1 PSMG1 2 K7EK45 Polypyrimidine tract-binding protein 1 PTBP1 2 Q15365 Poly(rC)-binding protein 1; Poly(rC)-binding protein 3 PCBP1; 2 PCBP3 Q9BSJ8 Extended synaptotagmin-1 ESYT1 2 H3BP35 Diphosphomevalonate decarboxylase MVD 2 H0Y8C4 Serine/threonine-protein phosphatase 2A 56 kDa regulatory subunit delta isoform PPP2R5D 2 K7EP09 Bifunctional coenzyme A synthase; Phosphopantetheine adenylyltransferase; Dephospho- COASY 2 CoA kinase F8VNT9 Tetraspanin; CD63 antigen CD63 2 Q8NEV1 Casein kinase II subunit alpha 3; Casein kinase II subunit alpha CSNK2A3; 2 CSNK2A1 F8VTQ5 Heterogeneous nuclear ribonucleoprotein A1 HNRNPA1 2 A0A087X2E2 Carcinoembryonic

antigen-related cell adhesion molecule 8; Carcinoembryonic antigen- CEACAM8; 2 related cell adhesion molecule 1; Carcinoembryonic antigen-related cell adhesion CEACAM5; molecule 5 CEACAM1 H0YAS8 Clusterin; Clusterin beta chain; Clusterin alpha chain; Clusterin CLU 2 E9PP54 Tubulin-specific chaperone cofactor E-like protein TBCEL 2 P43034 Platelet-activating factor acetylhydrolase IB subunit alpha PAFAH1B1 2 Q9NT62 Ubiquitin-like-conjugating enzyme ATG3 ATG3 2 K7ERZ3 Perilipin-3 PLIN3 2 Q6DD88 Atlastin-3 ATL3 2 Q15084-3 Protein disulfide-isomerase A6 PDIA6 2 K7EQH4 ATP synthase subunit alpha, mitochondrial ATP5A1 2 G3V1U5 Vesicle transport protein GOT1B GOLT1B 2 O00186 Syntaxin-binding protein 3 STXBP3 2 Q5T6W2 Heterogeneous nuclear ribonucleoprotein K HNRNPK 2 H0YEEY4 ADP-sugar pyrophosphatase NUDT5 2 P35580 Myosin-10 MYH10 2 F5H081 Solute carrier family 2, facilitated glucose transporter member 4 SLC2A4 2 Q9H1C7 Cysteine-rich and transmembrane domain-containing protein 1 CYSTM1 2 K7EJ83 Cyclin-dependent kinase 2; Cyclin-dependent kinase 3 CDK3; CDK2 2 C9J352 Importin subunit alpha-5; Importin subunit alpha-5, N-terminally processed KPNA1 2 H9KV75 Alpha-actinin-1; Alpha-actinin-4; Alpha-actinin-2; Alpha-actinin-3 ACTN1; ACTN4; 2 ACTN3; ACTN2 A0A0A0MQS1 Pyrroline-5-carboxylate reductase; Pyrroline-5-carboxylate reductase 3 PYCRL 2 D6RBY0 Rieske domain-containing protein RFESD 2 C9JC71 Low affinity immunoglobulin gamma Fc region receptor III-A FCGR3A 2 P51811 Membrane transport protein XK XK 2 C9J1G2 DnaJ homolog subfamily B member 2 DNAJB2 2 Q9UN37 Vacuolar protein sorting-associated protein 4A; Vacuolar protein sorting-associated protein VPS4A; VPS4B; 2 4B; Fidgetin-like protein 1 FIGNL1 P61160 Actin-related protein 2 ACTR2 2 P16930 Fumarylacetoacetase FAH 2 A0A087X0K4 CUB and sushi domain-containing protein 2 CSMD2 2 P10746 Uroporphyrinogen-III synthase UROS 2 K7EQ02 DAZ-associated protein 1 DAZAP1 2 F5GYN4 Ubiquitin thioesterase OTUB1 OTUB1 2 P10599 Thioredoxin TXN 2 O95197 Reticulon-3 RTN3 2 E7ETB3 Aspartyl aminopeptidase DNPEP 2 P14868 Aspartate--tRNA ligase, cytoplasmic DARS 2 Q08AM6 Protein VAC14 homolog VAC14 2 Q93034 Cullin-5 CUL5 2 Q9BQA1 Methylosome protein 50 WDR77 2 A0A0B4J2G9 Ubiquitin-conjugating enzyme E2 L3 UBE2L3 2 Q04446 1,4-alpha-glucan-branching enzyme GBE1 2 Q96NA2 Rab-interacting lysosomal protein RILP 2 Q92539 Phosphatidate phosphatase LPIN2 LPIN2 2 P28482 Mitogen-activated protein kinase 1 MAPK1 2 Q7Z406 Myosin-14; Myosin-11 MYH14; 2 11 Q8IU68 Transmembrane channel-like protein 8 TMC8 2 Q96BJ3 Axin interactor, dorsalization-associated protein AIDA 2 P20042 Eukaryotic translation initiation factor 2 subunit 2 EIF2S2 2 A0A0A0MQR0 Docosahexaenoic acid omega-hydroxylase CYP4F3; Phylloquinone omega-hydroxylase CYP4F2; CYP4F3; 2 CYP4F2; Cytochrome P450 4F12; Phylloquinone omega-hydroxylase CYP4F11 CYP4F11; CYP4F12 Q9BS26 Endoplasmic reticulum resident protein 44 ERP44 2 O00178 GTP-binding protein 1 GTPBP1 2 P27797 Calreticulin CALR 2 G3V0E5 Transferrin receptor protein 1; Transferrin receptor protein 1, serum form TFRC 2 P48147 Prolyl endopeptidase PREP 2 H7BZC1 Hippocalcin-like protein 1; Neuron-specific calcium-binding protein hippocalcin; Neurocalcin- HPCAL1; 2 delta NCALD; HPCA F8VPD4 CAD protein; Glutamine-dependent carbamoyl-phosphate synthase; Aspartate CAD 2 carbamoyltransferase; Dihydroorotase Q9NTJ5 Phosphatidylinositol phosphatase SAC1 SACM1L 2 P26447 Protein S100-A4 S100A4 2 Q9UIW2 Plexin-A1 PLXNA1 2 H0YJS0 V-type proton ATPase subunit D ATP6V1D 2 P25445 Tumor necrosis factor receptor superfamily member 6 FAS 2 C9JEU5 Fibrinogen gamma chain FGG 2 F5H562 Copper-transporting ATPase 2; WND/140 kDa ATP7B 2 B5MCF3 Protein GUCD1 GUCD1 2 Q96TA1 Niban-like protein 1 FAM129B 2 F5GWT4 Serine/threonine-protein kinase WNK1 WNK1 2 E9PLT1 Platelet glycoprotein 4 CD36 2 Q8TDB8 Solute carrier family 2, facilitated glucose transporter member 14; Solute carrier family 2, SLC2A14; 2 facilitated glucose transporter member 3 SLC2A3 B4DDD6 Drebrin-like protein DBNL 2 E9PII3 Band 4.1-like protein 2 EPB41L2 2 E9PNF5 Glutamine-dependent NAD(+) synthetase NADSYN1 2 K7ELL7 Glucosidase 2 subunit beta PRKCSH 2 Q9BV20 Methylthioribose-1 -phosphate isomerase MRI1 2 J3KNB4 Cathelicidin antimicrobial peptide;

Antibacterial protein FALL-39; Antibacterial protein LL-37 CAMP 2 Q5T1Z0 Phospholysine phosphohistidine inorganic pyrophosphate phosphatase LHPP 2 H3BLV0 Complement decay-accelerating factor CD55 2 Q9Y3E7 Charged multivesicular body protein 3 CHMP3 2 Q5T6H7 Xaa-Pro aminopeptidase 1 XPNPEP1 2 O94779 Contactin-5 CNTN5 2 E9PNR2 Ras and Rab interactor 1 RIN1 2 E7ESJ7 Protein FAM114A2 FAM114A2 2 A0A0A0MS99 Multidrug resistance-associated protein 1 ABCC1 2 A0A0G2JM15 Large neutral amino acids transporter small subunit 4 SLC43A2 2 H0YBF7 Arf-GAP with SH3 domain, ANK repeat and PH domain-containing protein 1; Arf-GAP with ASAP1; 2 SH3 domain, ANK repeat and PH domain-containing protein 2 ASAP2 P29992 Guanine nucleotide-binding protein subunit alpha-11 GNA11 2 F8WF69 Clathrin light chain A CLTA 2 B5MCN0 Atlastin-2 ATL2 2 F8W9F9 Serine/threonine-protein kinase WNK2; Serine/threonine-protein kinase WNK3 WNK2; WNK3 2 A2A3F3 Transient receptor potential cation channel subfamily M member 3 TRPM3 2 Q5KU26 Collectin-12 COLEC12 2 O14523 C2 domain-containing protein 2-like C2CD2L 2 Q96DG6 Carboxymethylenebutenolidase homolog CMBL 2 Q5THJ4 Vacuolar protein sorting-associated protein 13D VPS13D 2 Q02790 Peptidyl-prolyl cis-trans isomerase FKBP4; Peptidyl-prolyl cis-trans isomerase FKBP4, N- FKBP4 2 terminally processed A0A0G2JQD2 Glutathione S-transferase theta-1 GSTT1 2 Q9NV96 Cell cycle control protein 50A; Cell cycle control protein 50B TMEM30A; 2 TMEM30B P62330 ADP-ribosylation factor 6 ARF6 2 Q9UPU5 Ubiquitin carboxyl-terminal hydrolase 24 USP24 2 B2R4S9 Histone H2B HIST1H2BI 2 P10599 Thioredoxin TXN 2 R4GN98 Protein S100; Protein S100-A6 S100A6 2 H3BS66 Small integral membrane protein 1 SMIM1 2 Q8NHG7 Small VCP/p97-interacting protein SVIP 2 C9JIG9 Serine/threonine-protein kinase OSR1; STE20/SPS1-related proline-alanine-rich protein OXSR1; 2 kinase STK39 Q5VZR0 Golgi-associated plant pathogenesis-related protein 1 GLIPR2 2 F8WD49 Anion exchange protein 3; Anion exchange protein; Anion exchange protein 2 SLC4A3; 2 SLC4A2 P61626 Lysozyme C; Lysozyme LYZ 2 Q5T123 SH3 domain-binding glutamic acid-rich-like protein 3 SH3BGRL3 2 F5H571 Ubiquitin carboxyl-terminal hydrolase 5 USP5 2 Q9H3K6 BclA-like protein 2 BOLA2; 2 BOLA2B P05164 Myeloperoxidase; Myeloperoxidase; 89 kDa myeloperoxidase; 84 kDa MPO 2 myeloperoxidase; Myeloperoxidase light chain; Myeloperoxidase heavy chain P50395 Rab GDP dissociation inhibitor beta GDI2 2 G3V2U7 Acylphosphatase; Acylphosphatase-1 ACYP1 2 J3KNB4 Cathelicidin antimicrobial peptide; Antibacterial protein FALL-39; Antibacterial protein LL-37 CAMP 2 H0Y9X3 Programmed cell death protein 6 PDCD6 2 A6NJA2 Ubiquitin carboxyl-terminal hydrolase; Ubiquitin carboxyl-terminal hydrolase 14 USP14 2 P35998 26S protease regulatory subunit 7 PSMC2 2 P30041 Peroxiredoxin-6 PRDX6 2 P35612 Beta-adducin ADD2 2 P00918 Carbonic anhydrase 2 CA2 2 A0A0C4DGH5 Cullin-associated NEDD8-dissociated protein 1 CAND1 2 C9JZN1 Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-2; Guanine nucleotide- GNB2; 1 binding protein subunit beta-4 GNB4 I3L4X8 Integrin beta; Integrin beta-3 ITGB3 1 H7BZJ3 PDIA3 1 P17066 Heat shock 70 kDa protein 6; Putative heat shock 70 kDa protein 7 HSPA6; 1 HSPA7 J3KT70 Arf-GAP with dual PH domain-containing protein 2 ADAP2; 1 CENTA2 Q9Y6M5 Zinc transporter 1 SLC30A1 1 P54725 UV excision repair protein RAD23 homolog A RAD23A 1 F8WD59 40S ribosomal protein SA RPSA; 1 RPSAP58 P06753 1 E9PMI6 Methylosome subunit pICln CLNS1A 1 Q9BVK6 Transmembrane emp24 domain-containing protein 9 TMED9 1 C9JJV6 Myeloid-associated differentiation marker MYADM 1 Q8IXQ3 Uncharacterized protein C9orf40 C9orf40 1 Q8NDC0 MAPK-interacting and spindle-stabilizing protein-like MAPK1IP1L 1 Q99747 Gamma-soluble NSF attachment protein NAPG 1 E7EWE1 Ubiquitin-like modifier-activating enzyme 5 UBA5 1 J3KNE3 Platelet-activating factor acetylhydrolase IB subunit beta PAFAH1B2 1 Q8IUI8 Cytokine receptor-like factor 3 CRLF3 1 Q9BTX7 Alpha-tocopherol transfer protein-like TTPAL 1 P63261 Actin, cytoplasmic 2; Actin, cytoplasmic 2, N-terminally processed ACTG1 1 P31153 S-adenosylmethionine synthase isoform type-2 MAT2A 1 O00560 Syntenin-1 SDCBP 1 Q4VB86 Protein 4.1 EPB41 1 Q8WUD1 Ras-related protein Rab-2B

RAB2B; 1 DKFZp313C1541 Q9NWX4 UPF0587 protein C1orf123 C1orf123 1 Q9Y2Z0
 Suppressor of G2 allele of SKP1 homolog SUGT1 1 H0YK48 Tropomyosin alpha-1 chain TPM1 1
 H3BS66 Small integral membrane protein 1 SMIM1 1 Q5VTS0 Neurensin-1 NRSN1 1 H0Y904
 Multidrug resistance-associated protein 7 ABCC10 1 P08F94 Fibrocystin PKHD1 1 G3V5X4
 Nesprin-2 SYNE2 1 P48507 Glutamate--cysteine ligase regulatory subunit GCLM 1 H3BUF4
 Cyclin-D1-binding protein 1 CCNDBP1 1 F8WB30 Target of Myb protein 1 TOM1 1 E5RJI8 CA1
 1 S4R3E5 Importin subunit alpha-7 KPNA6 1 E7EVS6 ACTB 1 P04206 Ig kappa chain V-III
 region GOL; Ig kappa chain V-III region WOL; Ig kappa chain V-III region 1 Ti; Ig kappa chain V-
 III region SIE Q5SSV3 N(G),N(G)-dimethylarginine dimethylaminohydrolase 2 DDAH2 1
 H0YG54 Oligoribonuclease, mitochondrial REXO2 1 P08246 Neutrophil elastase ELANE 1
 P40926 Malate dehydrogenase, mitochondrial MDH2 1 P98172 Ephrin-B1 EFNB1 1 Q96DD7
 Protein shisa-4 SHISA4 1 Q71RC9 Small integral membrane protein 5 SMIM5 1 P68133 Actin,
 alpha skeletal muscle; Actin, alpha cardiac muscle 1; Actin, gamma-enteric smooth ACTA1;
 ACTC1; 1 muscle; Actin, aortic smooth muscle ACTG2; ACTA2 H0Y9Q6 Clathrin light chain B
 CLTB 1 G3V5P0 KTN1 1 Q9NWX6 Probable tRNA(His) guanylyltransferase THG1L 1 Q15404
 Ras suppressor protein 1 RSU1 1 B2R4S9 Histone H2B HIST1H2B 1 K7EK06 Phenylalanine--
 tRNA ligase alpha subunit FARSA 1 H7C3S9 COP9 signalosome complex subunit 8 COPS8 1
 J3KRV4 Dual specificity mitogen-activated protein kinase kinase 3 MAP2K3 1 Q15042 Rab3
 GTPase-activating protein catalytic subunit RAB3GAP1 1 P01893 Putative HLA class I
 histocompatibility antigen, alpha chain H; HLA class I histocompatibility HLA-H; HLA-C; 1
 antigen, Cw-6 alpha chain; HLA class I histocompatibility antigen, B-38 alpha chain; HLA HLA-B
 class I histocompatibility antigen, B-67 alpha chain; HLA class I histocompatibility antigen, B-82
 alpha chain; HLA class I histocompatibility antigen, B-39 alpha chain; HLA class I
 histocompatibility antigen, Cw-18 alpha chain; HLA class I histocompatibility antigen, Cw-7 alpha
 chain; HLA class I histocompatibility antigen, B-42 alpha chain; HLA class I histocompatibility
 antigen, B-14 alpha chain; HLA class I histocompatibility antigen, B-8 alpha chain; HLA class I
 histocompatibility antigen, B-7 alpha chain Q9UNW1 Multiple inositol polyphosphate phosphatase
 1 MINPP1 1 Q6B0K9 Hemoglobin subunit mu HBM 1 O75915 PRA1 family protein 3 ARL6IP5 1
 O95376 E3 ubiquitin-protein ligase ARIH2 ARIH2 1 E9PBW4 Hemoglobin subunit gamma-2
 HBG2 1 C9J1X0 WD repeat-containing protein 91 WDR91 1 Q8NCV1 1 Q9Y4P8 WD repeat
 domain phosphoinositide-interacting protein 2 WIPI2 1 A0A087WUX6 Proteasomal ubiquitin
 receptor ADRM1 ADRM1 1 Q9NQS7 Inner centromere protein INCENP 1 P62805 Histone H4
 HIST1H4A 1 Q8TDY2 RB1-inducible coiled-coil protein 1 RB1CC1 1 C9JFM5 Syntaxin-4 STX4
 1 B0QZ43 Erlin-2; Erlin-1 ERLIN1; 1 ERLIN2 C9J8T0 Selenocysteine-specific elongation factor
 EEFSEC 1 P61224 Ras-related protein Rap-1b; Ras-related protein Rap-1b-like protein RAP1B 1
 Q99828 Calcium and integrin-binding protein 1 CIB1 1 Q53GQ0 Very-long-chain 3-oxoacyl-CoA
 reductase HSD17B12 1 H0Y9Q9 ADP-ribosyl cyclase/cyclic ADP-ribose hydrolase 2 BST1 1
 R4GN98 Protein S100; Protein S100-A6 S100A6 1 Q8TB73 Protein NDNF NDNF 1 Q6P1A2
 Lysophospholipid acyltransferase 5 LPCAT3 1 F2Z2Y4 Pyridoxal kinase PDXK 1 Q9HA65 TBC1
 domain family member 17 TBC1D17 1 G3V126 V-type proton ATPase subunit H ATP6V1H 1
 D6RGE2 Isochorismatase domain-containing protein 1 ISOC1 1 A0A0B4J222 ADP-ribosylation
 factor-like protein 15 ARL15 1 F8W7W4 Androglobin ADGB 1 Q9GZR7 1 A0A087WVC4
 cAMP-dependent protein kinase catalytic subunit beta PRKACB 1 A8MU39 Serine/threonine-
 protein phosphatase; Serine/threonine-protein phosphatase 5 PPP5C 1 G3V1N2 HBA2 1 P69892
 Hemoglobin subunit gamma-2 HBG2 1 F8WEZ0 1-phosphatidylinositol 3-phosphate 5-kinase
 PIKFYVE 1 A0A087WZZ4 Ammonium transporter Rh type A RHAG 1 Q1JUQ3 Peptidyl-prolyl
 cis-trans isomerase; Peptidyl-prolyl cis-trans isomerase FKBP1A FKBP12- 1 Exin; FKBP1A
 F8VWZ5 H2.0-like homeobox protein HLX 1 P62877 E3 ubiquitin-protein ligase RBX1; E3
 ubiquitin-protein ligase RBX1, N-terminally processed RBX1 1 Q7Z5P9 Mucin-19 MUC19 1
 F8VPB3 TPK1 1 F8WBF4 Transmembrane protein 50B TMEM50B 1 C9JL85 Myotrophin MTPN

1 Q53EQ6 Tigger transposable element-derived protein 5 TIGD5 1 E3W974 DNPEP 1 H7BYV1 Interferon-induced transmembrane protein 1; Interferon-induced transmembrane protein IFITM2; IFITM3; 1 2; Interferon-induced transmembrane protein 3 IFITM1 D6RC06 Histidine triad nucleotide-binding protein 1 HINT1 1 E9PNW4 CD59 glycoprotein CD59 1 H7C2Z6 Grancalcin GCA 1 Q9H1C7 Cysteine-rich and transmembrane domain-containing protein 1 CYSTM1 1 H0YK07 ATP-dependent Clp protease ATP-binding subunit clpX-like, mitochondrial CLPX 1 E9PG15 14-3-3 protein theta YWHAQ 1 K7ENK9 Vesicle-associated membrane protein 3; Vesicle-associated membrane protein 2 VAMP2; VAMP3 1 F8WDD6 Cation channel sperm-associated protein subunit gamma CATSPERG 1 P35754 Glutaredoxin-1 GLRX 1 Q5THJ4 Vacuolar protein sorting-associated protein 13D VPS13D 1 B4E3H6 Transforming acidic coiled-coil-containing protein 1 TACC1 1 B0YJC4 Vimentin VIM 1 H7C5R6 AT-rich interactive domain-containing protein 4B ARID4B 1 P49755 Transmembrane emp24 domain-containing protein 10 TMED10 1 E9PMJ3 Ribonuclease inhibitor RNH1 1 B1AHA9 DNA replication licensing factor MCM5 MCM5 1 P23634 Plasma membrane calcium-transporting ATPase 4 ATP2B4 1 Q14152 Eukaryotic translation initiation factor 3 subunit A EIF3A 1 C9IY70 60S ribosomal export protein NMD3 NMD3 1 A0A087WUS7 Ig delta chain C region IGHD 1 P07384 Calpain-1 catalytic subunit CAPN1 1 G3V2C9 Guanine nucleotide-binding protein subunit gamma; Guanine nucleotide-binding protein GNG2 1 G(I)/G(S)/G(O) subunit gamma-2 Q9Y2Y8 Proteoglycan 3 PRG3 1 V9GY70 DNAJB12 1 U5GXS0 MAM and LDL-receptor class A domain-containing protein 1 MALRD1 1 Q92625 Ankyrin repeat and SAM domain-containing protein 1A ANKS1A 1 (89) The present description refers to a number of documents, the contents of which are herein incorporated by reference in their entirety.

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Claims

1. An in vitro method for preparing a clinical human blood sample, the method comprising (a) obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) from a blood sample of a subject having or suspected of having Parkinson's disease; (b) removing hemoglobin from the preparation of isolated EEV, thereby obtaining a processed blood sample depleted of hemoglobin; and (c) quantifying the expression levels of one or more proteins in the processed blood sample of (b).
2. The method of claim 1, wherein the quantification in (c) is performed on EEVs having a diameter of between 100 nm and 1000 nm.
3. The method of claim 1, wherein the EEV are CD235a+ extracellular vesicles.
4. The method of claim 1, wherein the EEV are TSG101+, Rabs+, CD9+, CD63+, CD81+, or any combination thereof.
5. The method of claim 1, wherein said blood sample is platelet-free plasma.
6. The method of claim 1, further comprising, prior to (a), inducing calcium-dependent production

- of EEV from activated erythrocytes in the blood sample of the subject using a calcium ionophore, cold-storage, or ATP-depletion.
7. The method of claim 1, wherein the preparation of isolated EEV is obtained by separating the EEV by flow cytometry, differential centrifugation, nanomembrane ultrafiltration, immunoabsorbent capture, size-exclusion chromatography, ultracentrifugation, magnetic activated cell sorting (MACS), nanoparticle tracking analysis, light scattering, electrophoretic light scattering, dynamic light scattering, electron microscopy, or any combination thereof.
8. The method of claim 1, wherein the quantification in (c) comprises mass spectrometry.
9. The method of claim 1, wherein the quantification in (c) comprises nano liquid chromatography tandem mass spectrometry (nanoLC MS/MS).
10. An in vitro method for preparing a clinical human blood sample, the method comprising (a) obtaining a preparation of isolated erythrocyte-derived extracellular vesicles (EEV) depleted of hemoglobin from a blood sample of a subject having or suspected of having Parkinson's disease; and (b) quantifying the expression levels of one or more proteins biomarkers, in the preparation of isolated EEV depleted of hemoglobin.
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